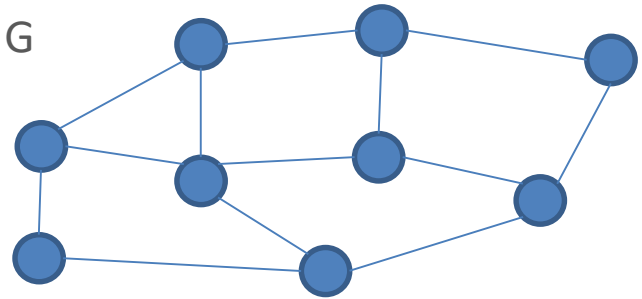
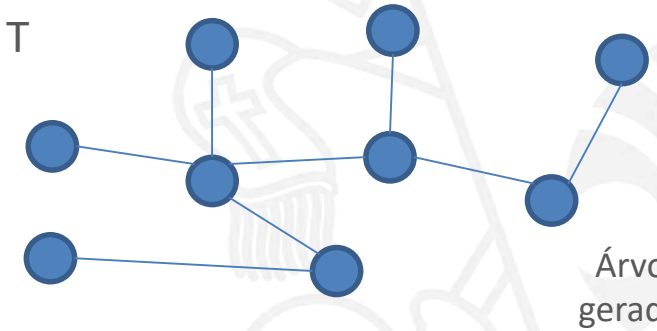
Árvore Geradora Mínima

Zenilton Patrocínio

Árvore Geradora – Lembrete

Uma árvore geradora $T = (V, E_T)$ de um grafo não direcionado $G = (V, E)$ é um subgrafo gerador de G que também é uma árvore, isto é , $E_T \subseteq E$, $|E_T| = |V| - 1$ e T é conexo.





Árvore geradora

Árvore Geradora de Custo Mínimo

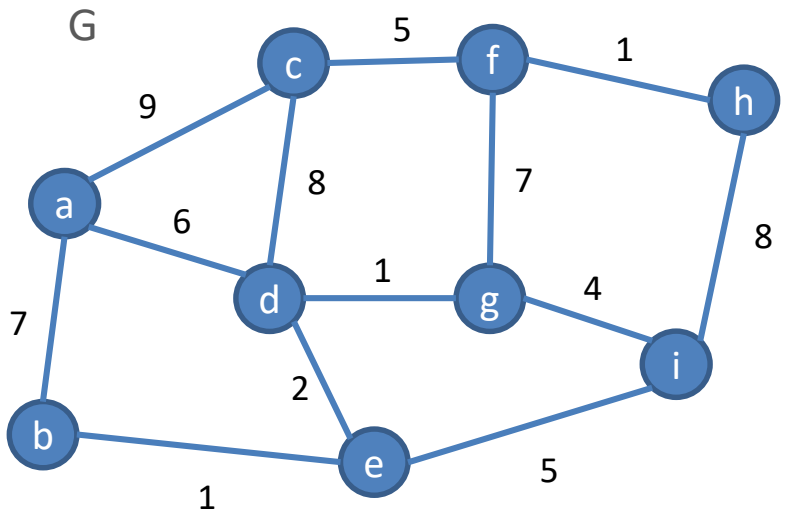
Seja um grafo não direcionado e ponderado G = (V, E) em que um valor de custo $c_e > 0$ é associado a cada aresta $e \in$ E.

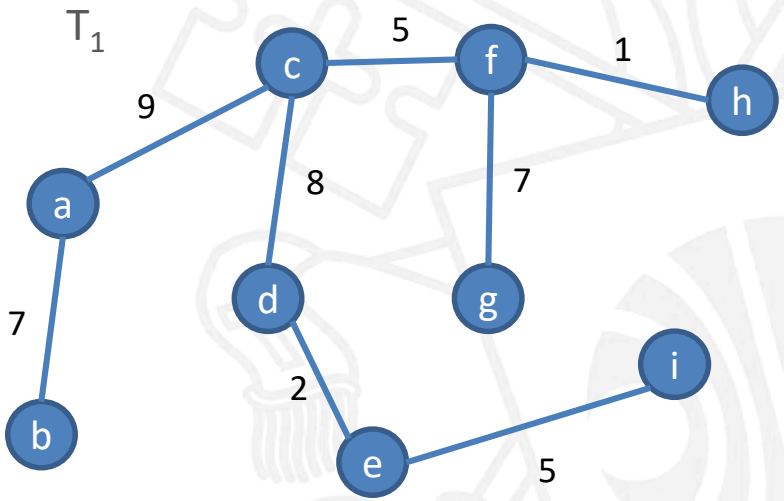
O custo total C(T) de uma árvore geradora T = (V, E_T) de G pode ser obtido pela soma dos custos de todas as arestas de T, isto é, $C(\mathrm{T}) = \sum_{e \in E_T} c_e$.

A árvore geradora de custo mínimo (AGM) é a árvore geradora de menor custo total dentre todas as possíveis árvores geradoras de um grafo.

A determinação de uma AGM pode ser feita em tempo polinomial.

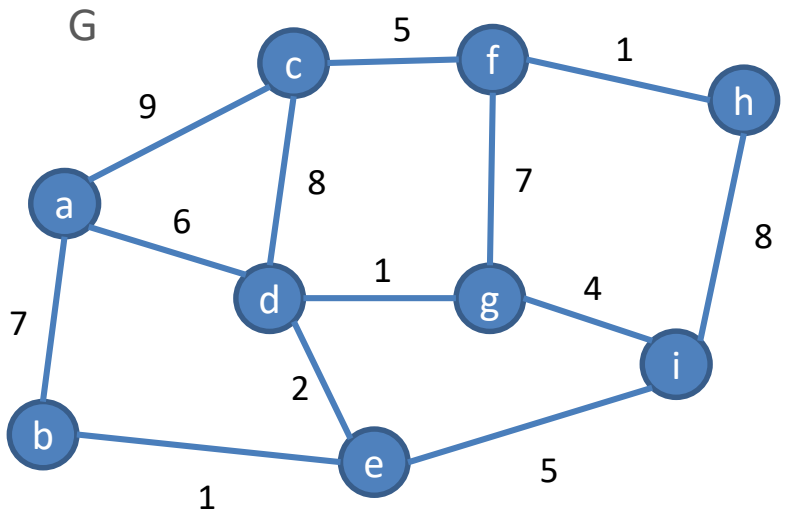
Árvore Geradora de Custo Mínimo

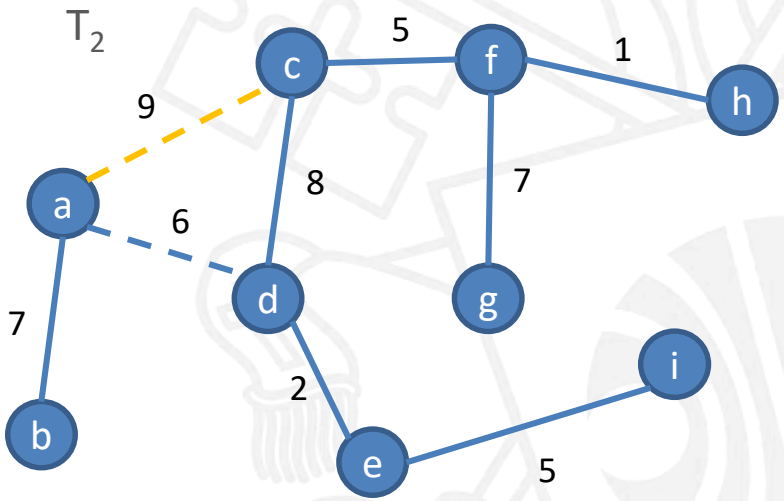




$C(T_1) = 44$

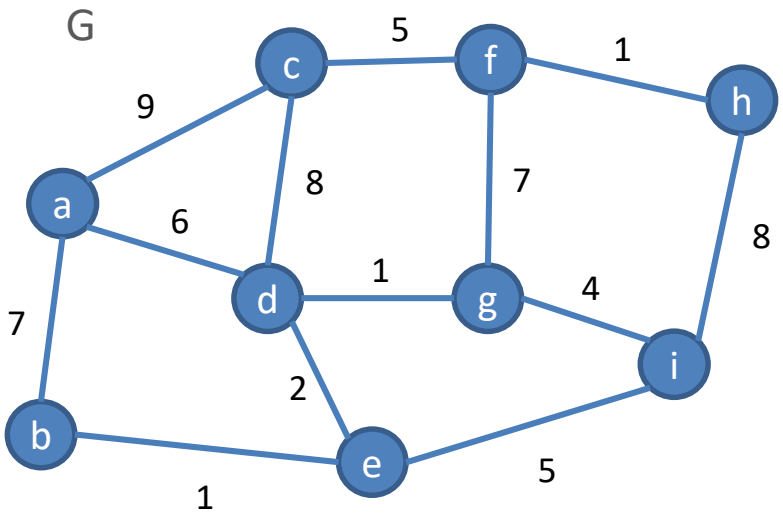
Árvore Geradora de Custo Mínimo

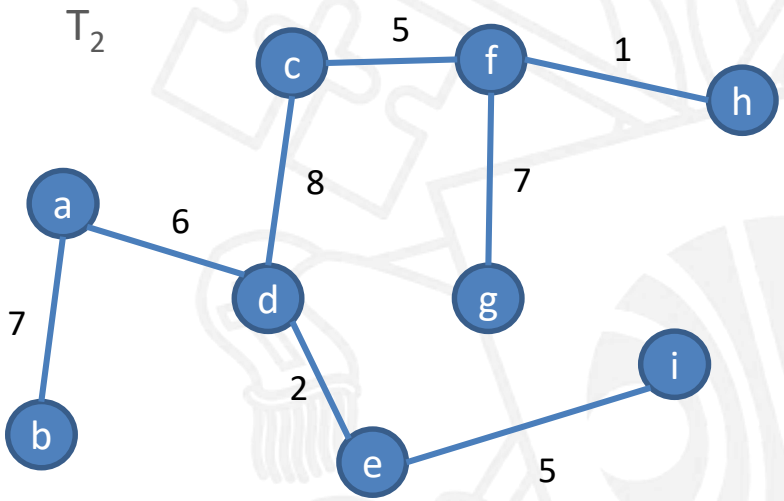




$C(T_2) = 41$

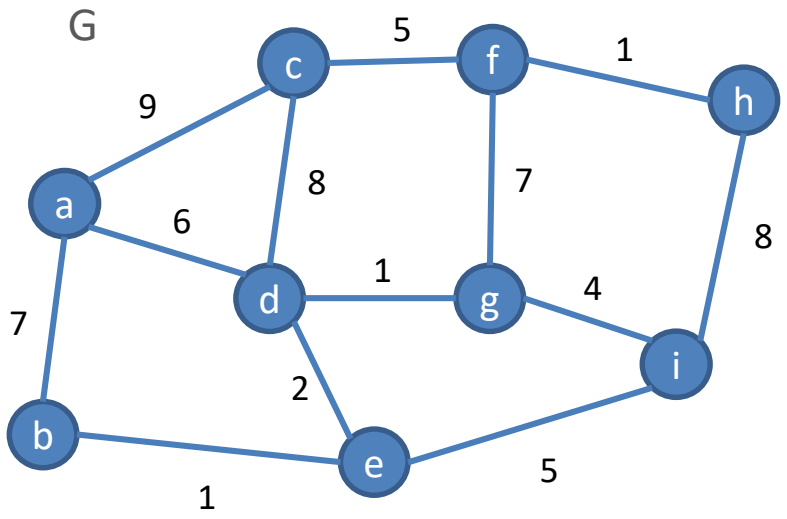
Árvore Geradora de Custo Mínimo

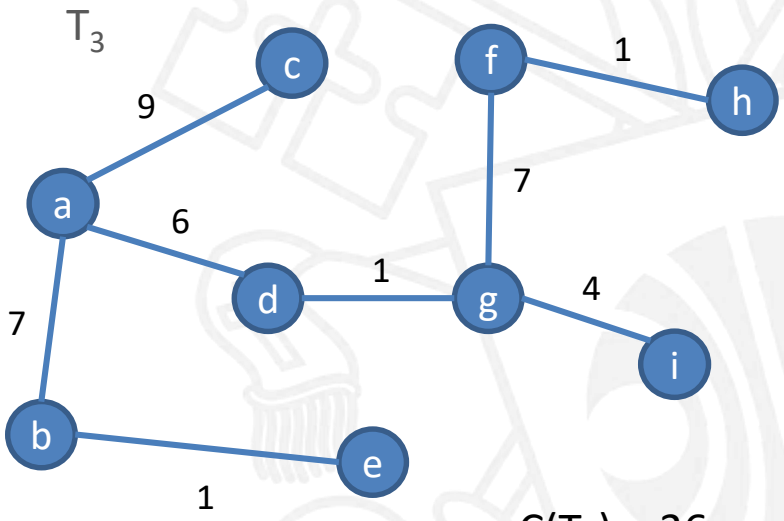




$C(T_2) = 41$

Árvore Geradora de Custo Mínimo





$C(T_3) = 36$

Método de Prim

Método de Prim

Este algoritmo foi proposto originalmente em 1930 por um matemático tcheco, posteriormente pelo americano Robert Prim em 1957 e redescoberto por Edsger Dijkstra em 1959.

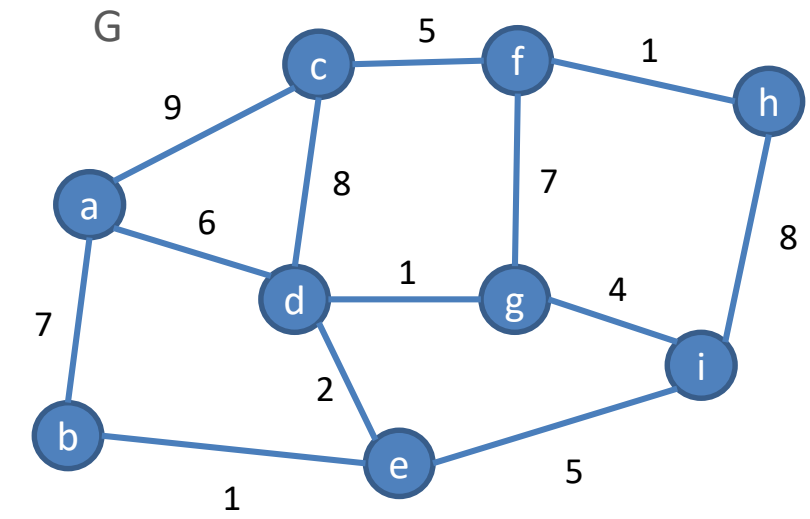
Incluir, de forma gulosa, um a um, os vértices da árvore geradora mínima.

A partir de um vértice qualquer do grafo (raiz), a cada passo, acrescenta-se a aresta de menor peso incidente ao conjunto de vértices que já foram selecionados e que possui uma extremidade em vértices no conjunto de não selecionados.

Método de Prim – Algoritmo

1. Escolher um vértice qualquer $r \in V(G)$ // Selecionar uma raiz
2. $V(T) \leftarrow \{ r \}$; // Inicializar conj. de vértices selecionados
3. $E(T) \leftarrow \varnothing$; // Inicializar conj. arestas da AGM
4. <u>enquanto</u> $V(T) \neq V(G)$ <u>efetuar</u> // Se houver vértice não selecionado?
   a. Encontrar a aresta $\{v, w\}$ de menor custo tal que $v \in V(T)$ e $w \notin V(T)$
      // Isto é, a aresta de menor custo entre selecionados e não selecionados
   b. Acrescentar w a $V(T)$ // Adicionar novo vértice à AGM
   c. Acrescentar $\{v, w\}$ a $E(T)$ // Adicionar nova aresta à AGM

Método de Prim – Exemplo



Vértice não explorado

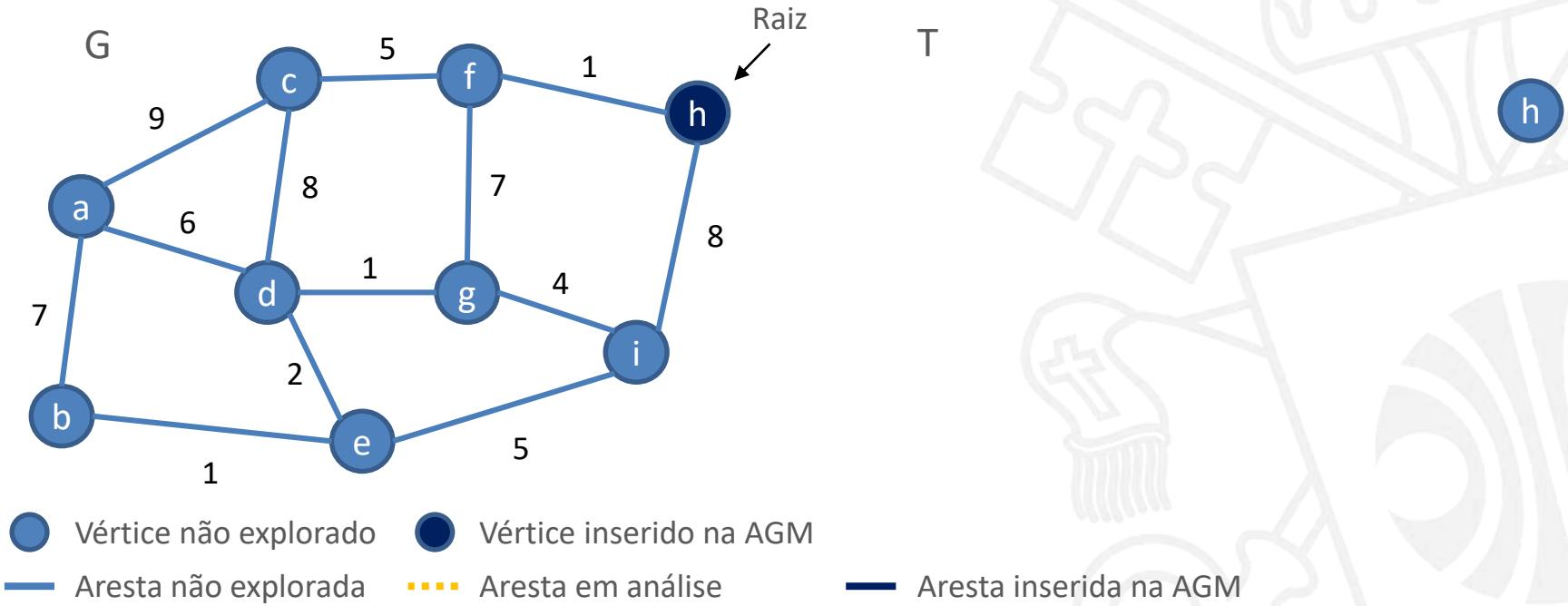
Vértice inserido na AGM

Aresta não explorada

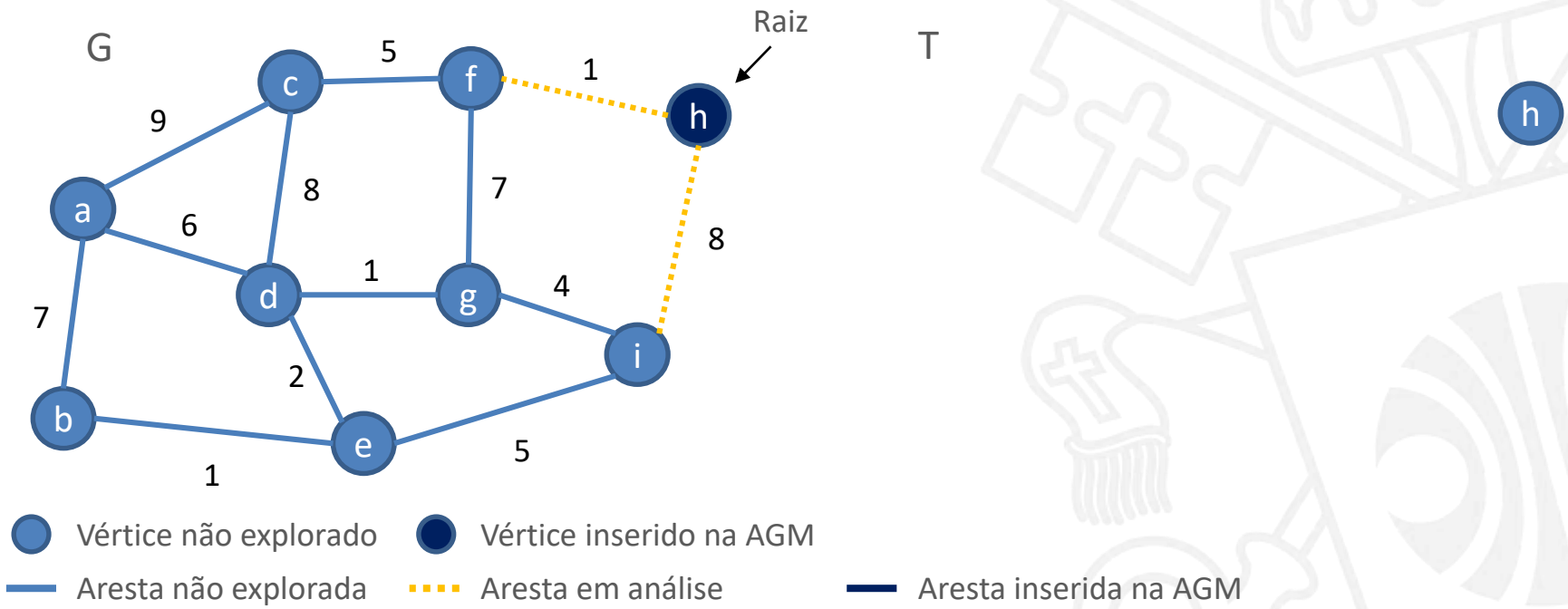
Aresta em análise

Aresta inserida na AGM

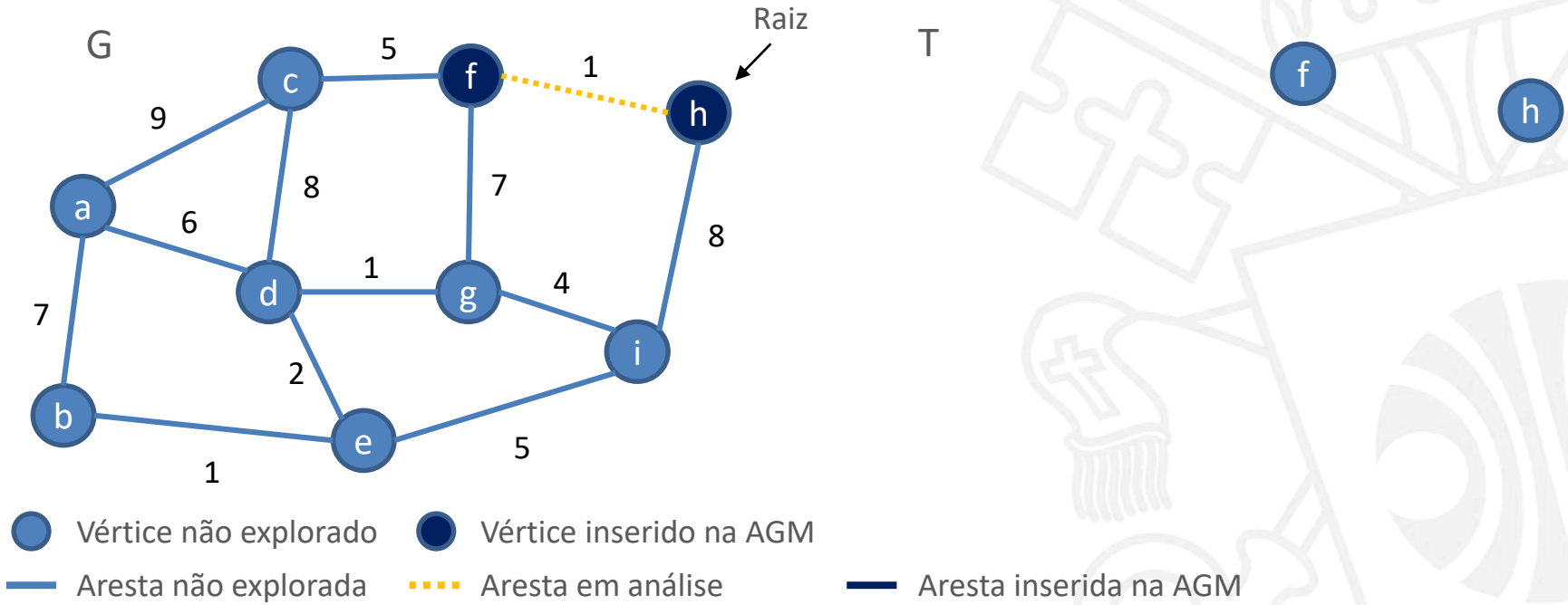
Método de Prim – Exemplo



Método de Prim – Exemplo



Método de Prim – Exemplo



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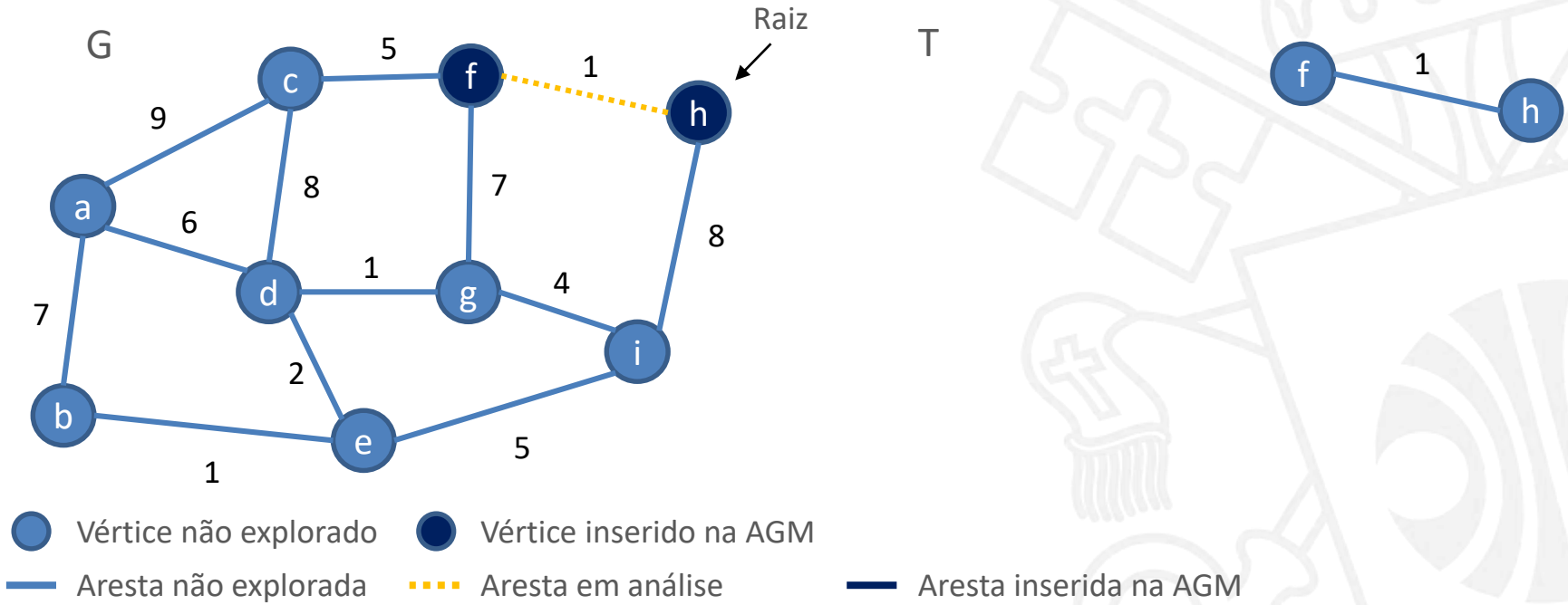

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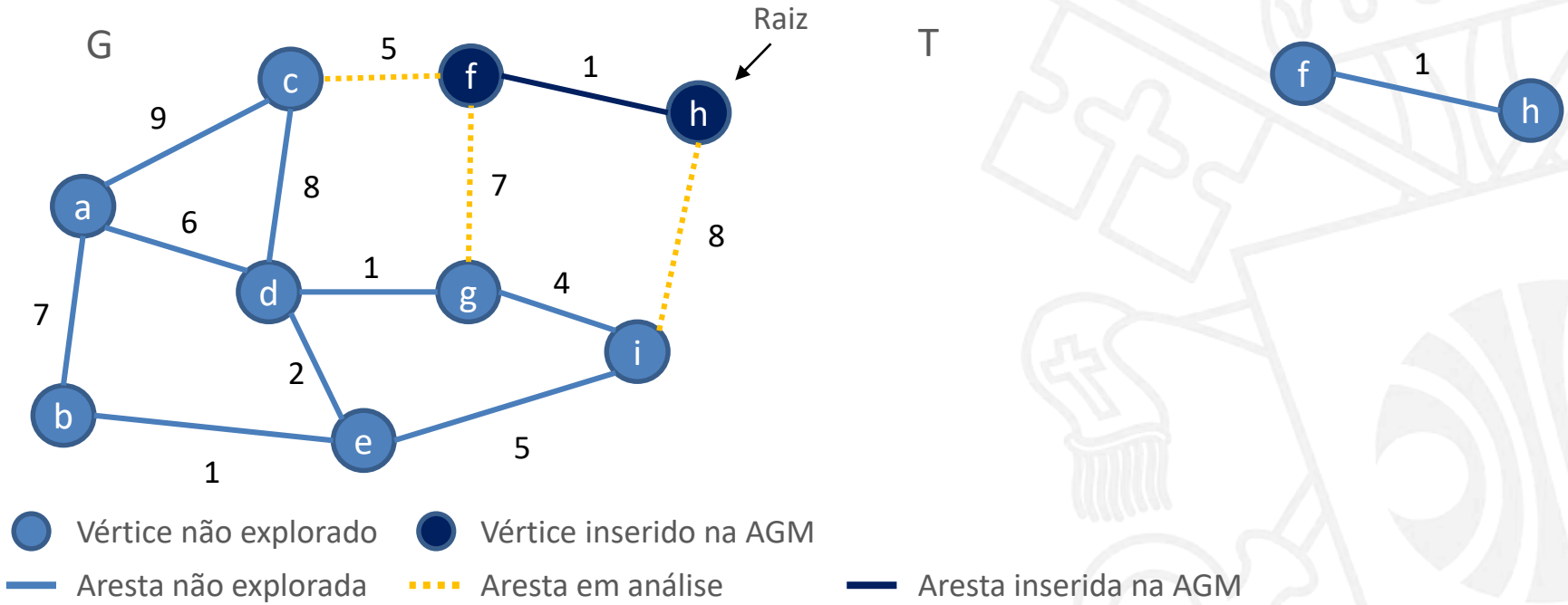

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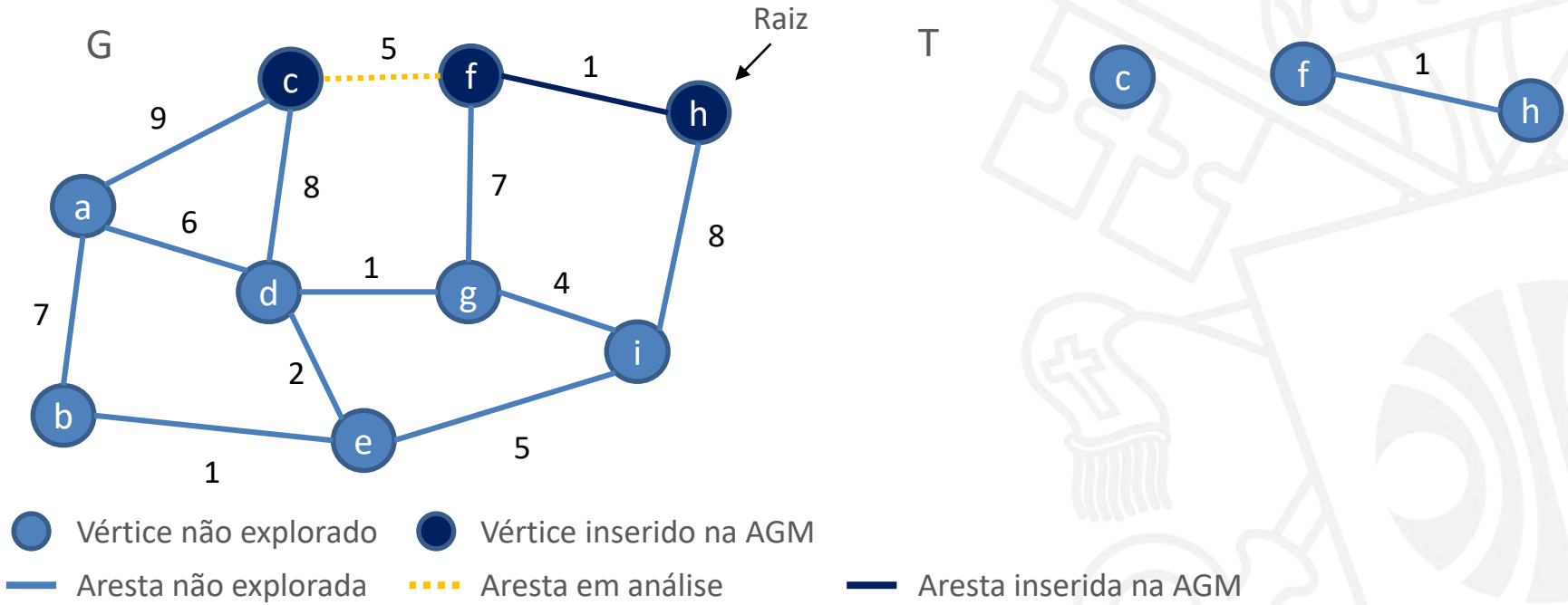

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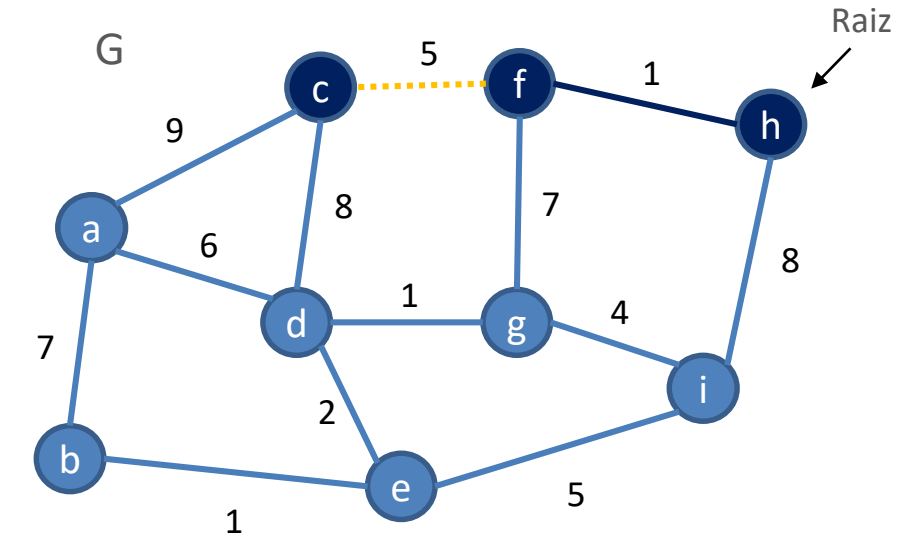

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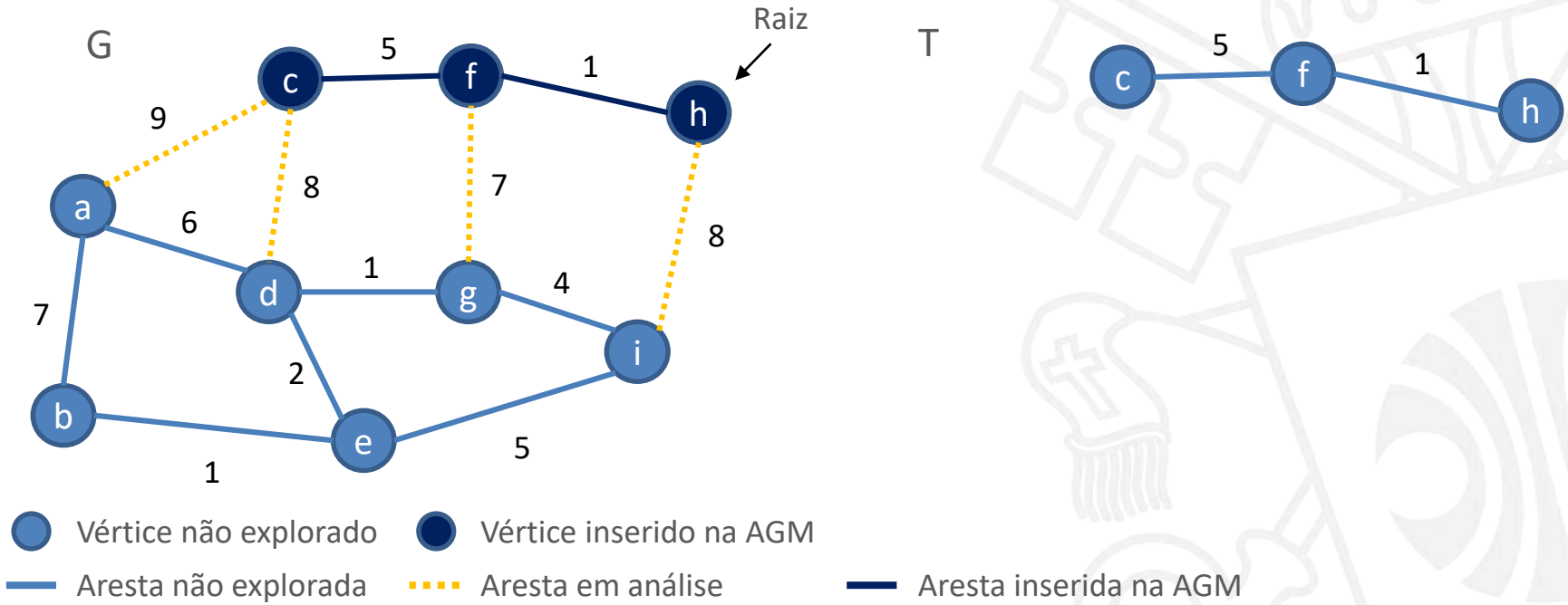

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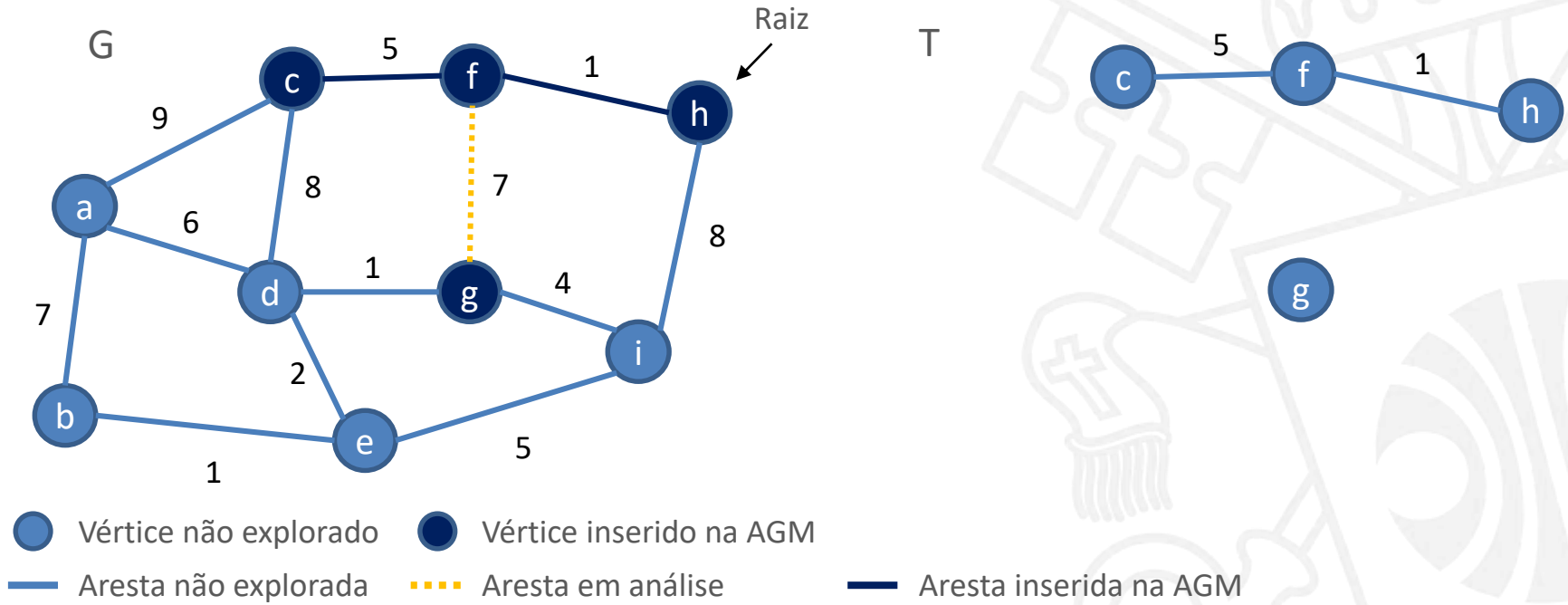

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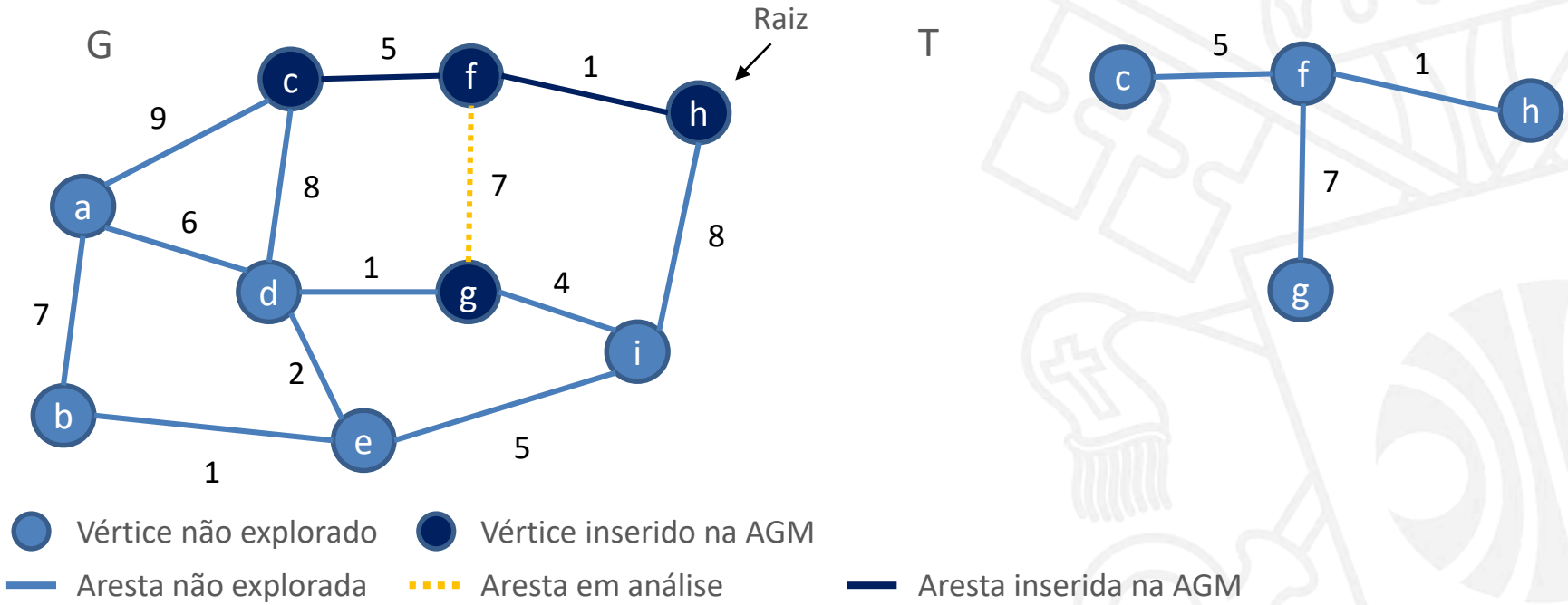

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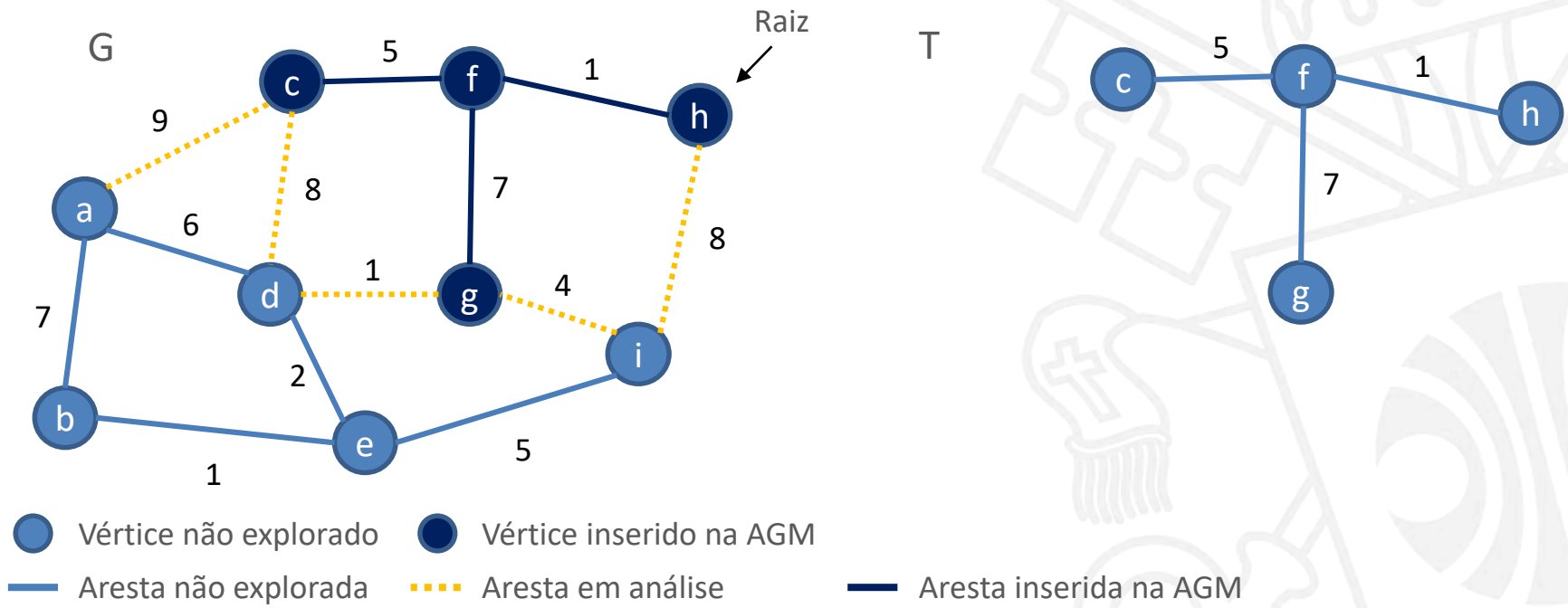

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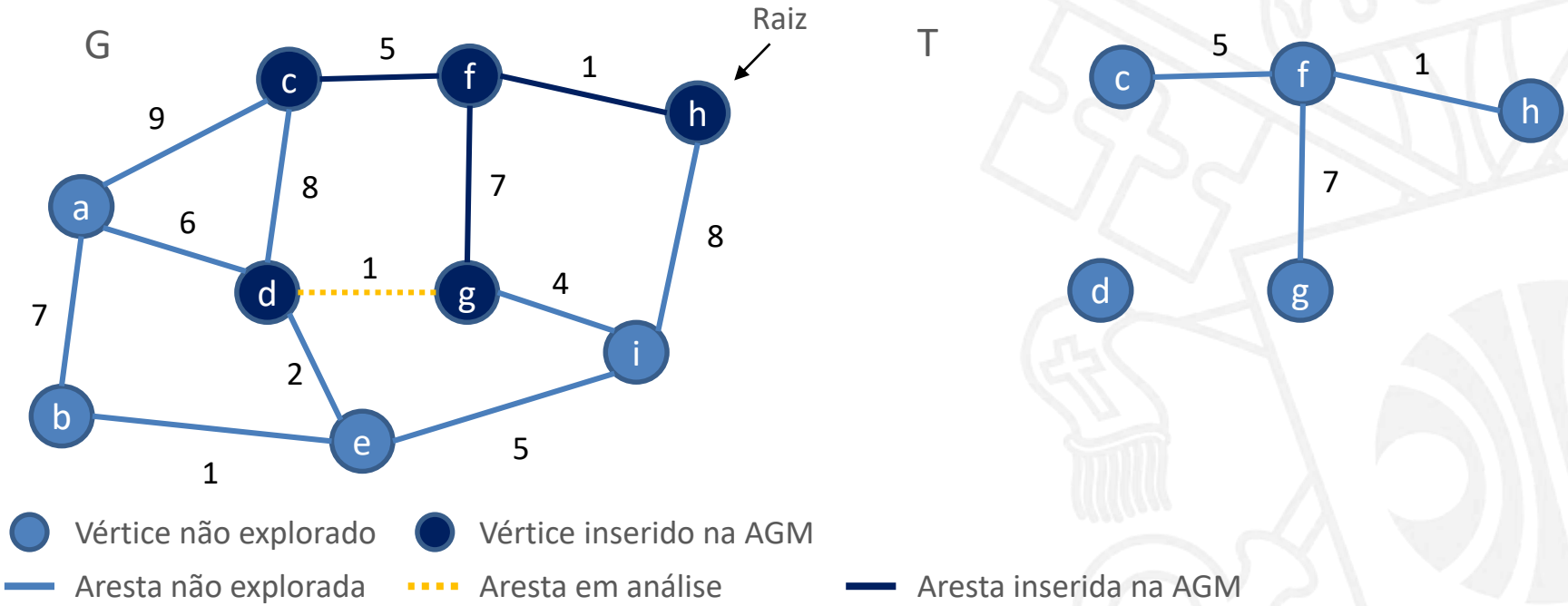

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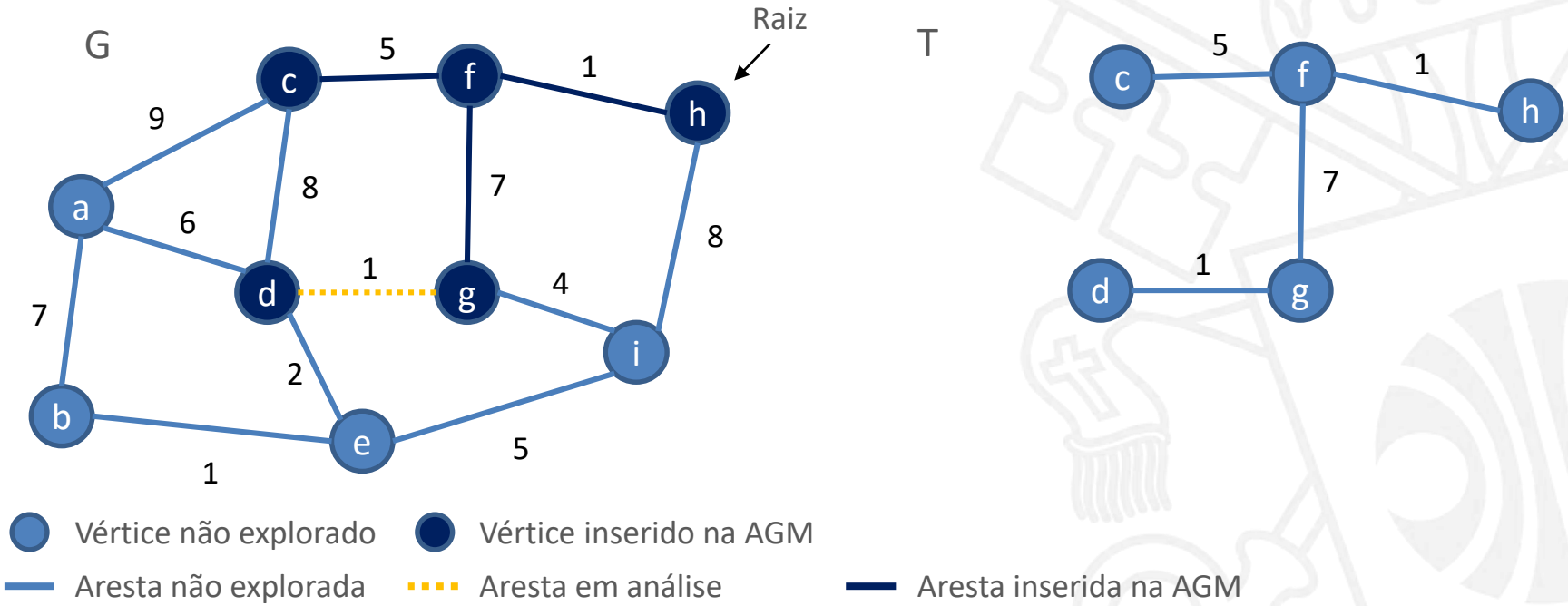

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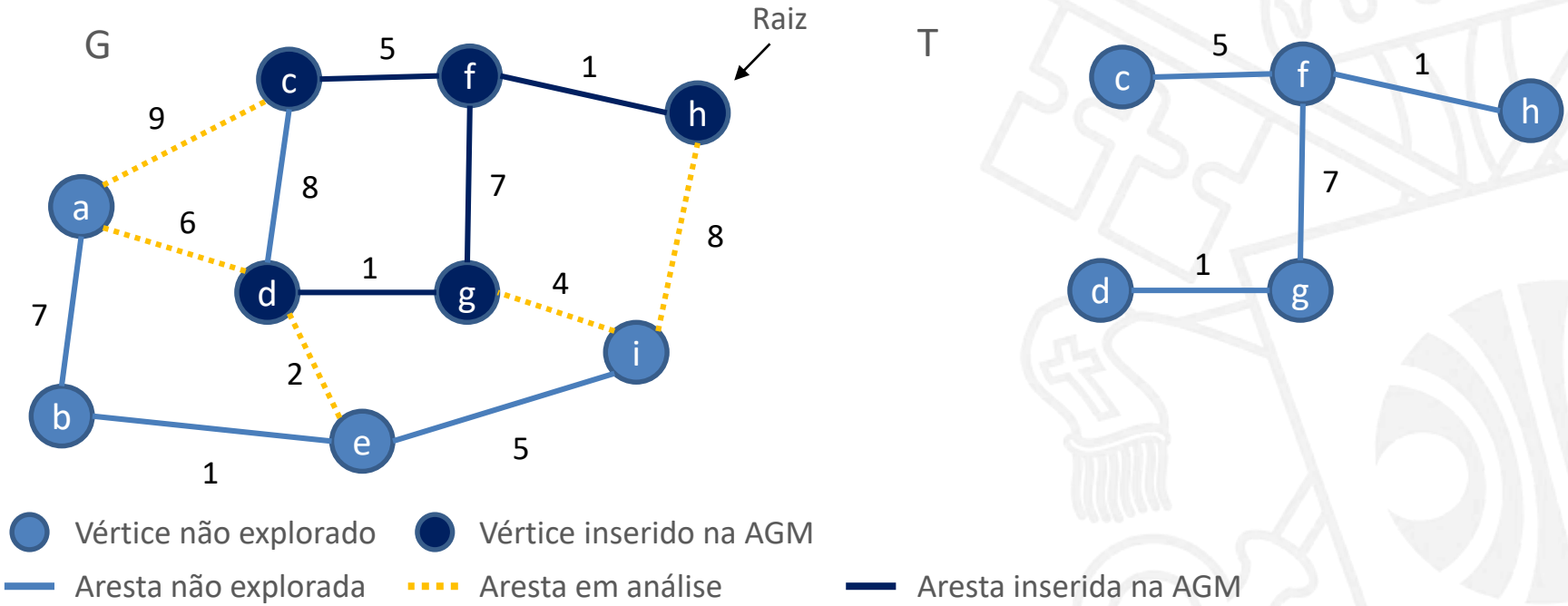

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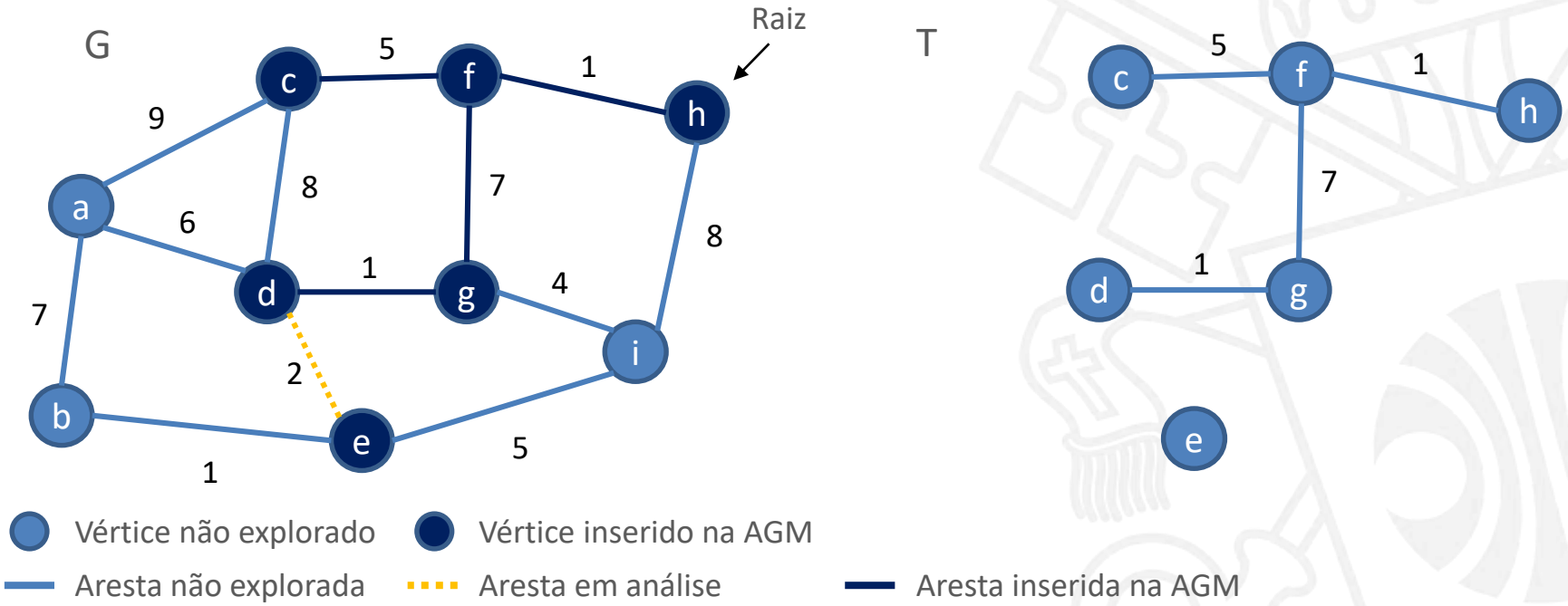

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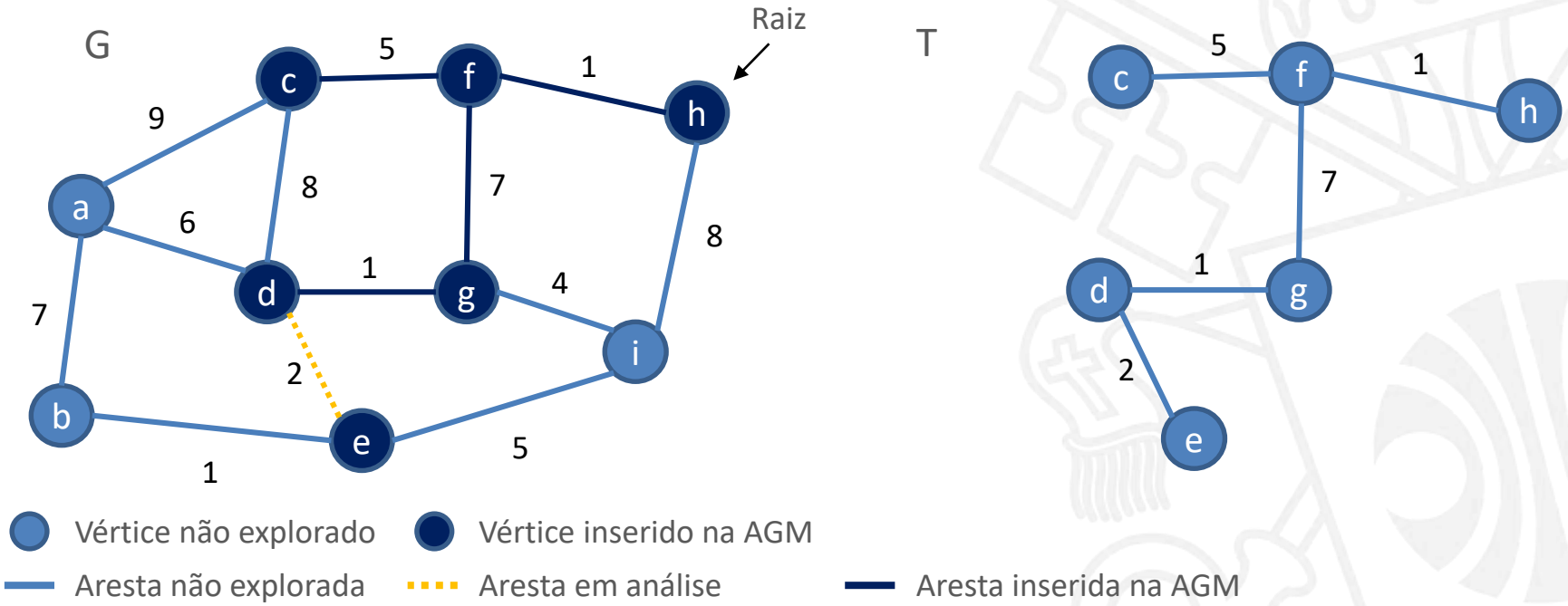

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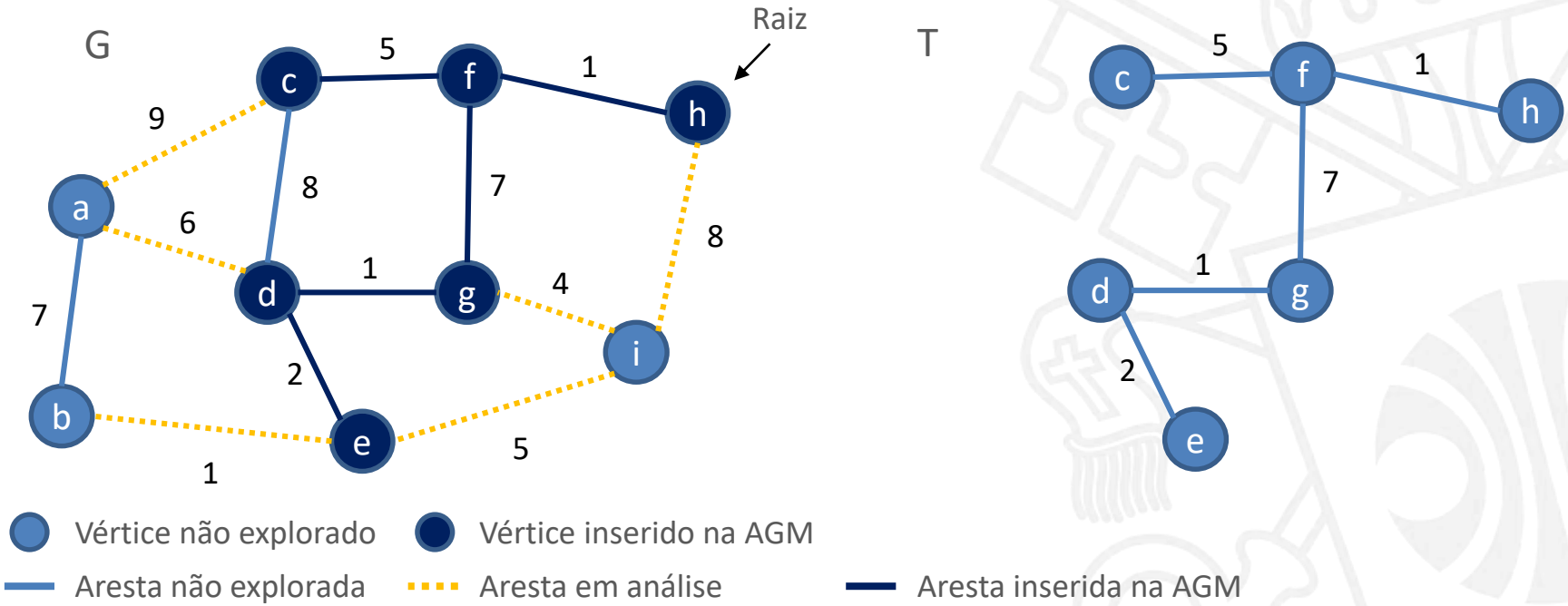

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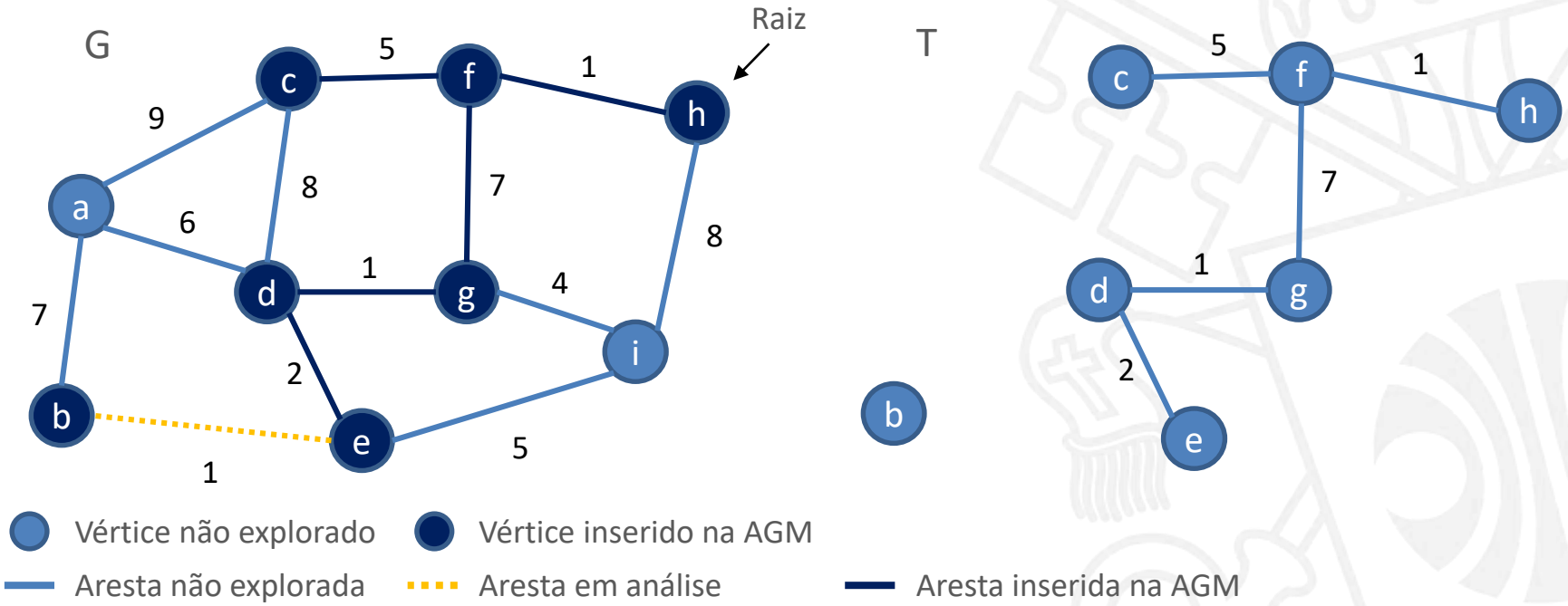

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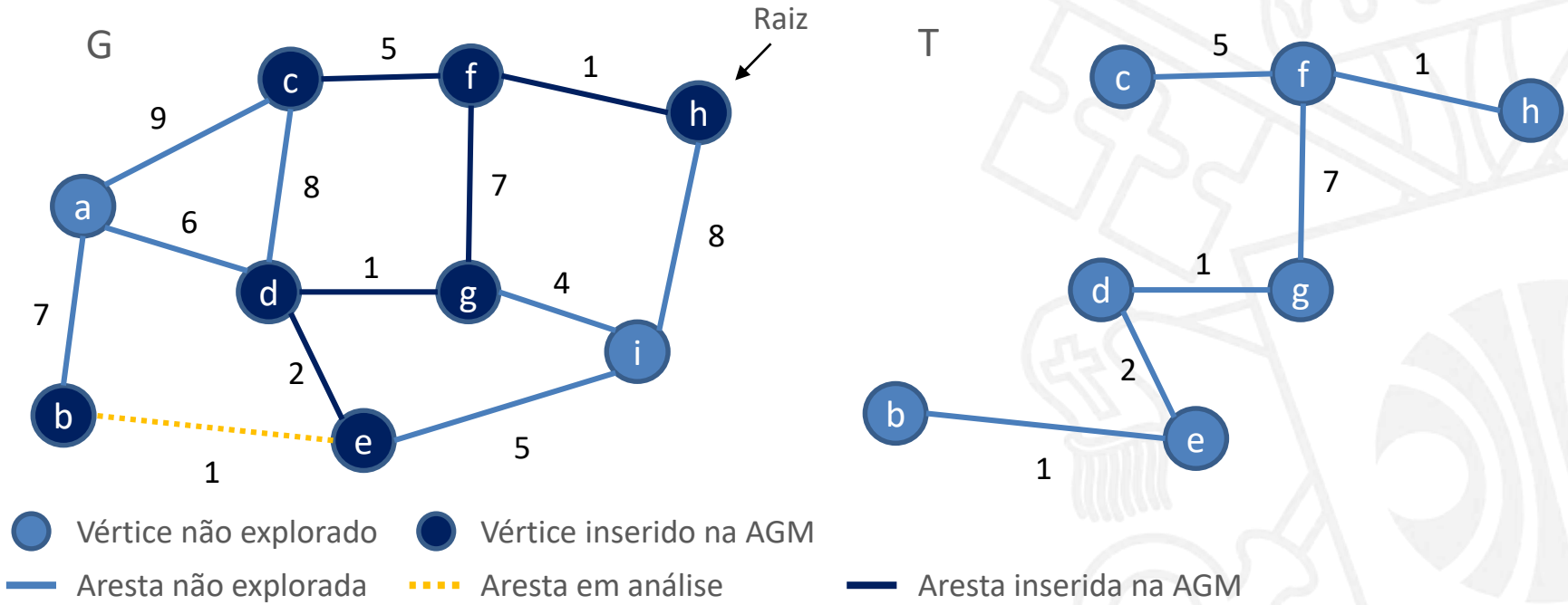

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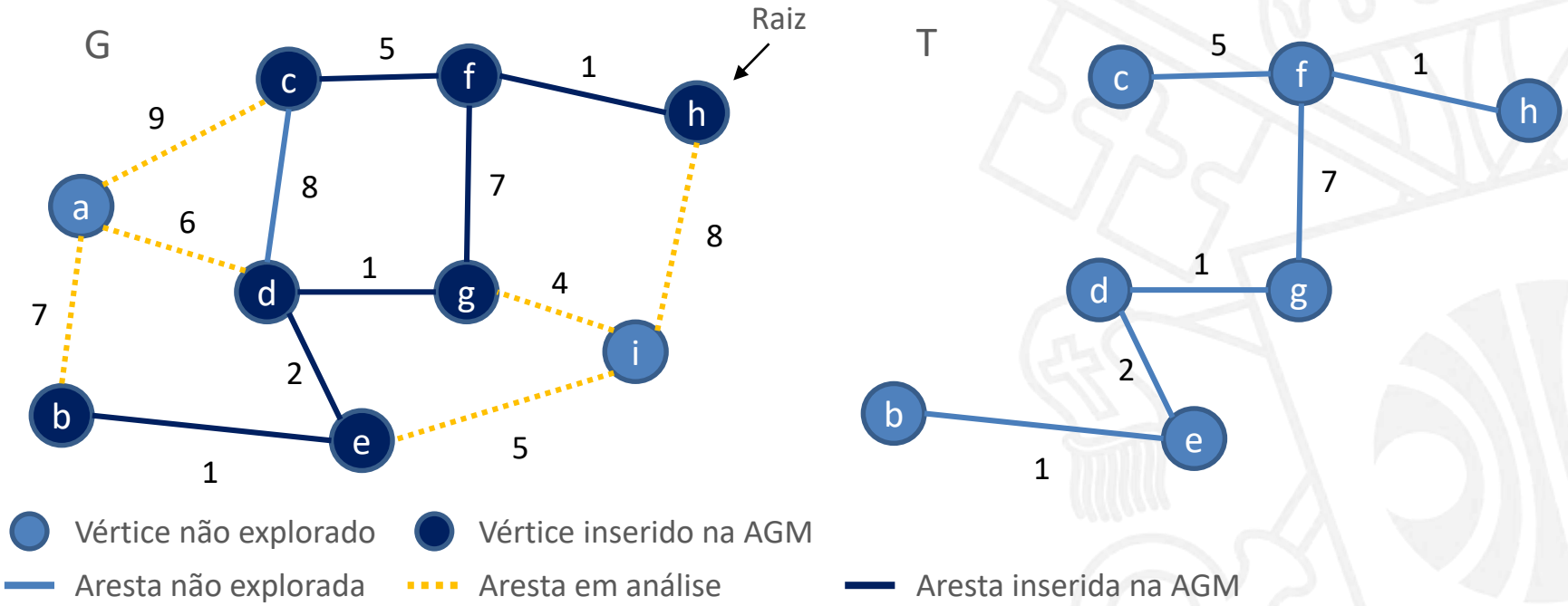

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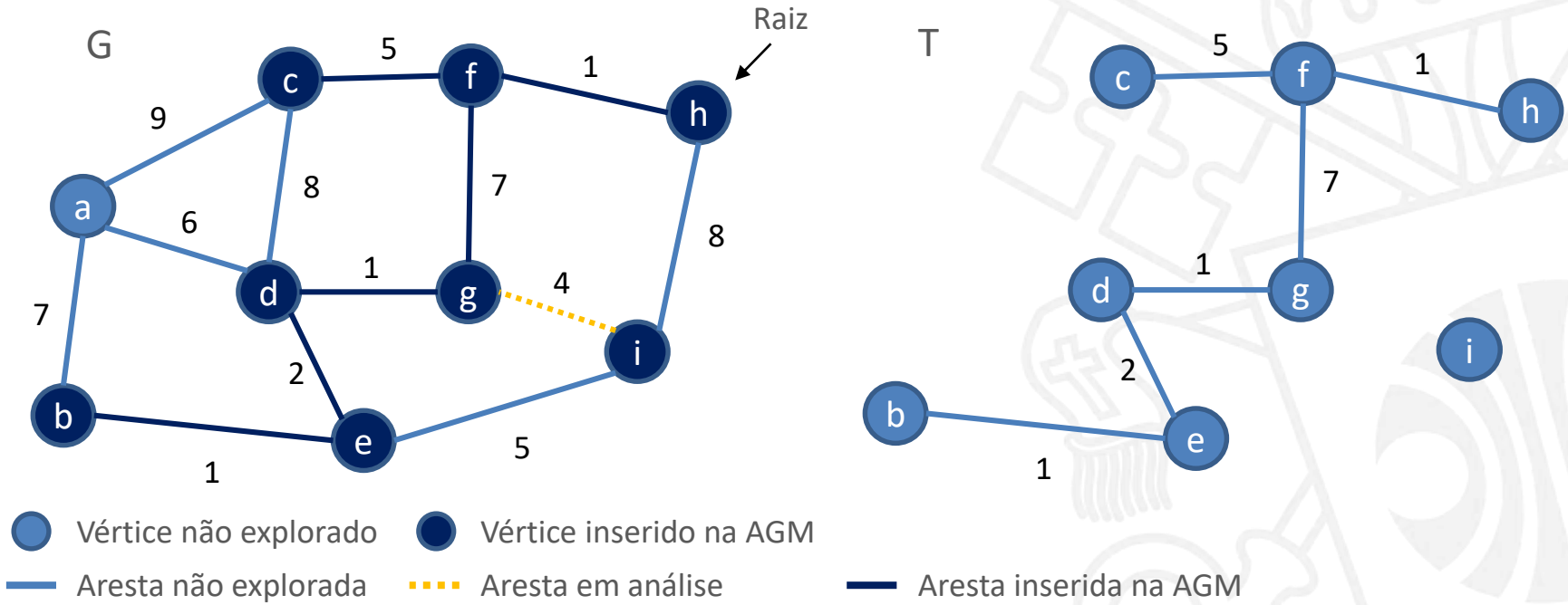

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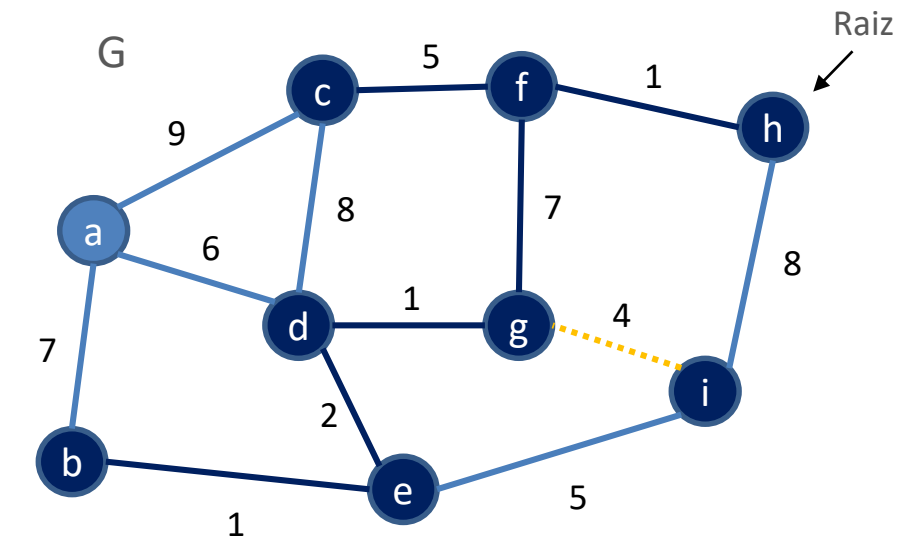

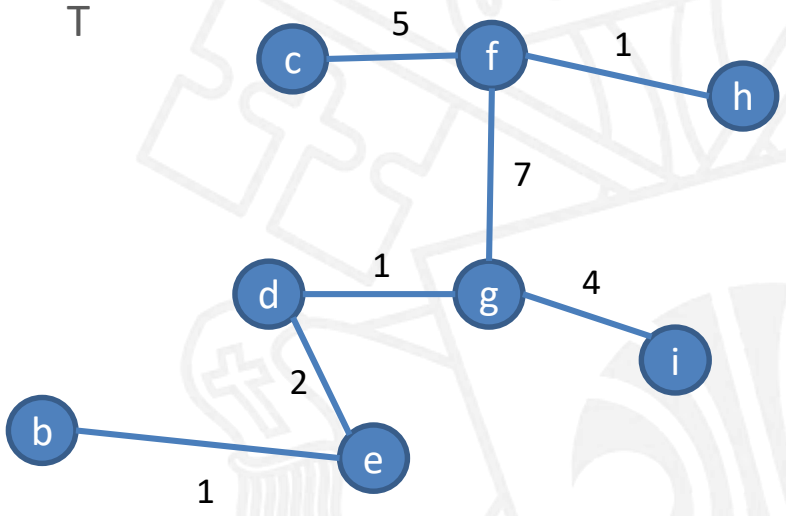

Vértice não explorado

Vértice inserido na AGM

Aresta não explorada

Aresta em análise

Aresta inserida na AGM

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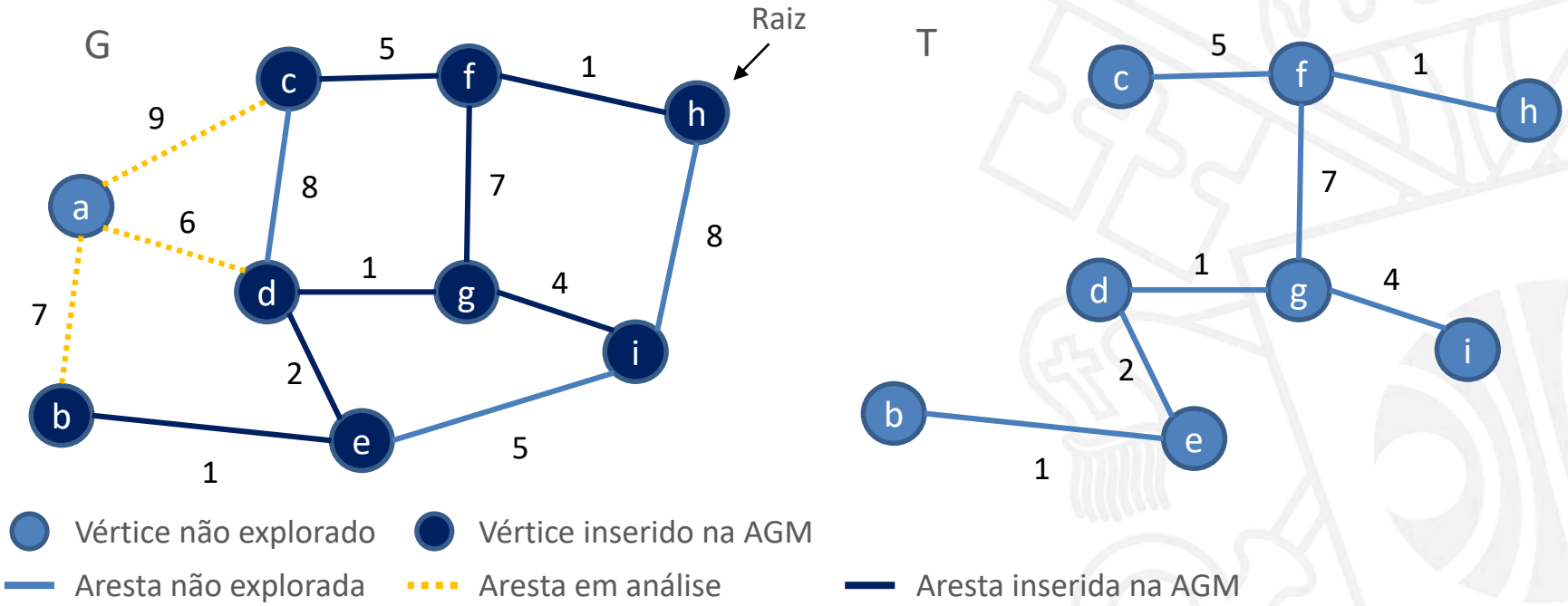

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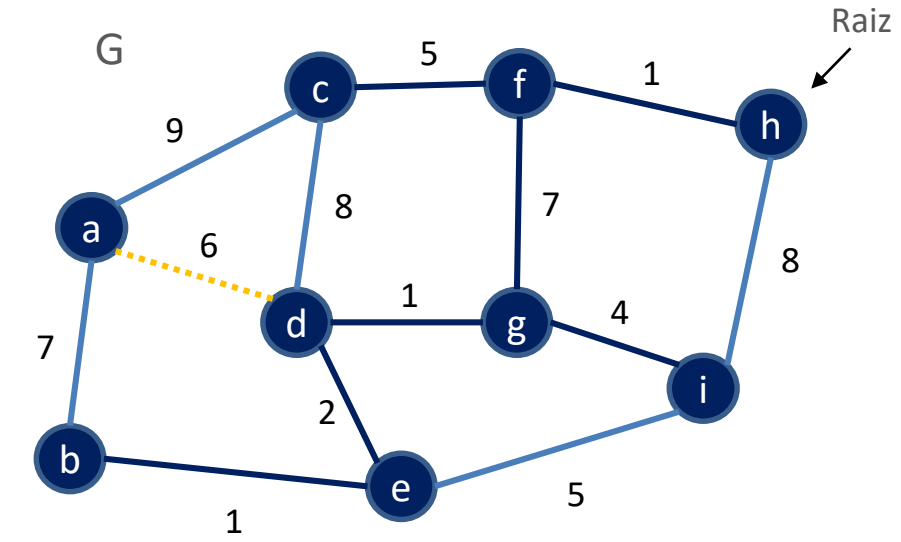

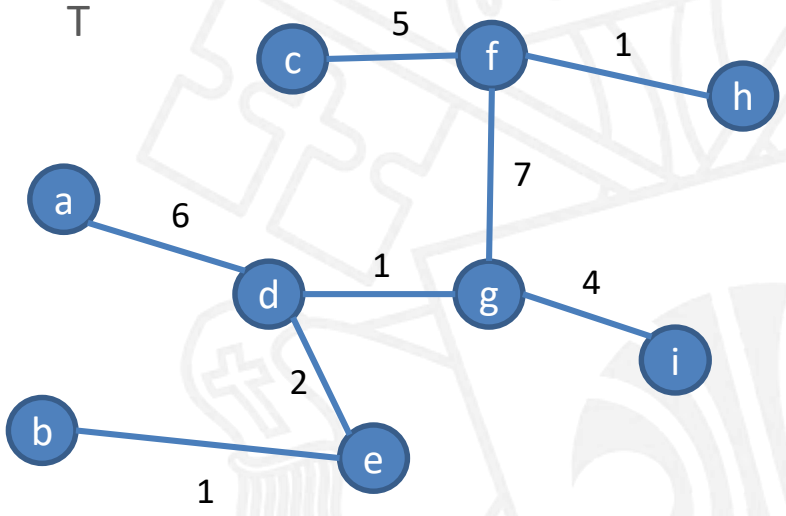

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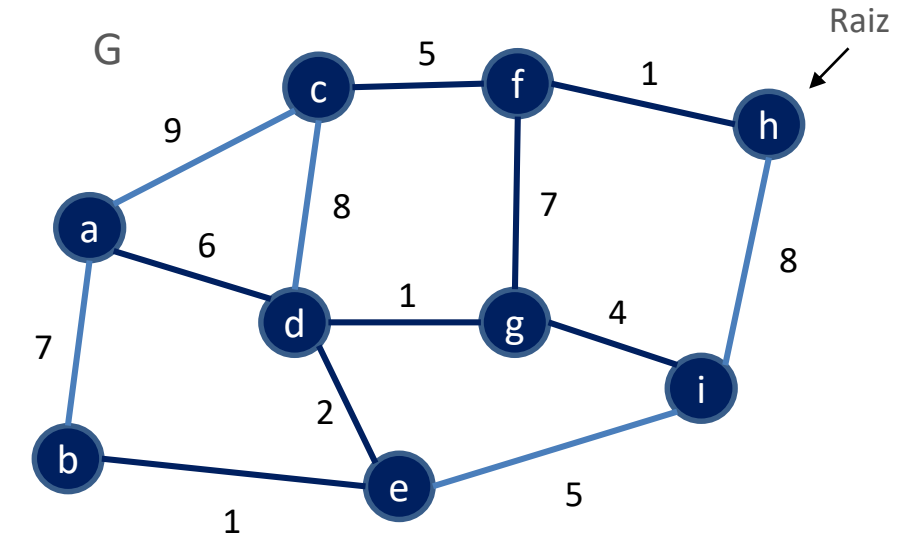

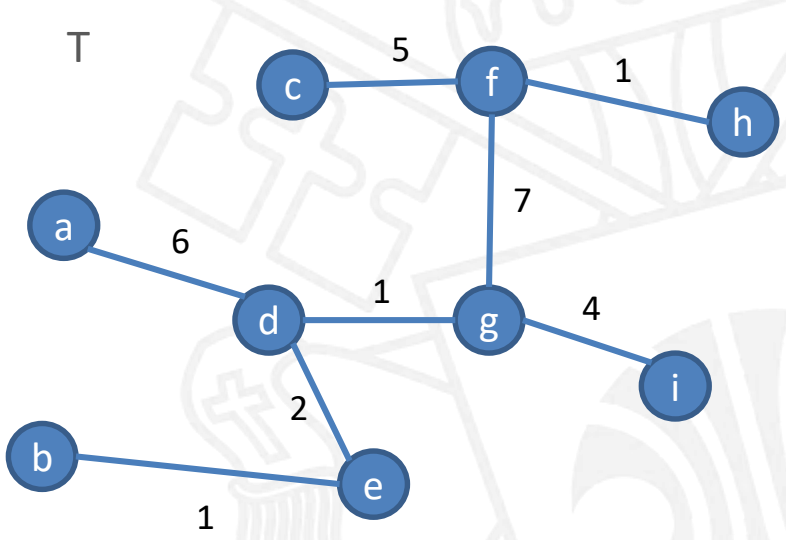

Vértice não explorado

Vértice inserido na AGM

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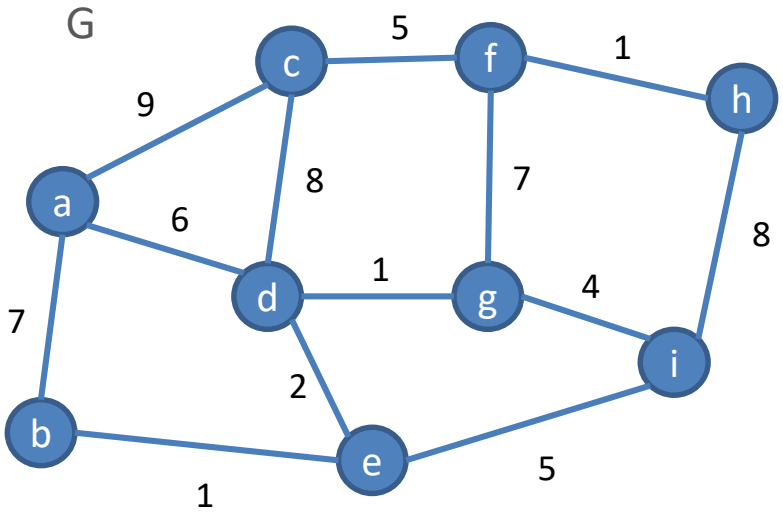
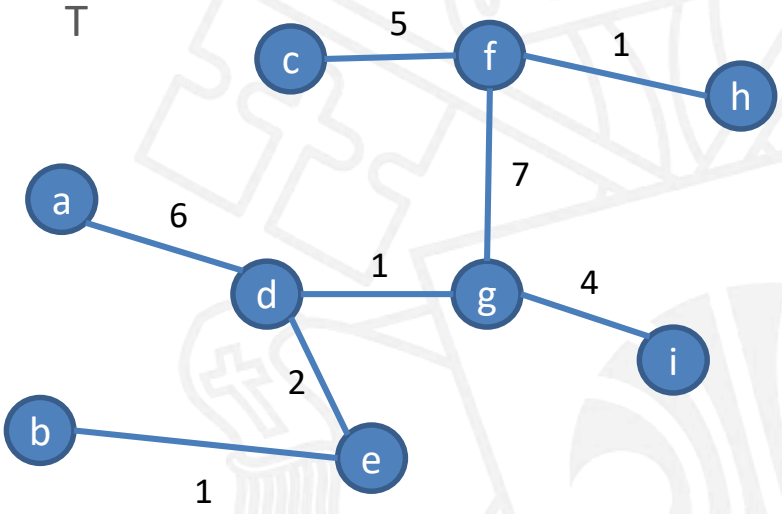
Método de Prim – Exemplo





AGM ⟹ C(T) = 27

Método de Kruskal

Método de Kruskal

Este algoritmo foi proposto por Kruskal em 1956 e ele propõe incluir na árvore, a cada iteração, a aresta de menor custo que **não** forme um ciclo com as inseridas anteriormente.

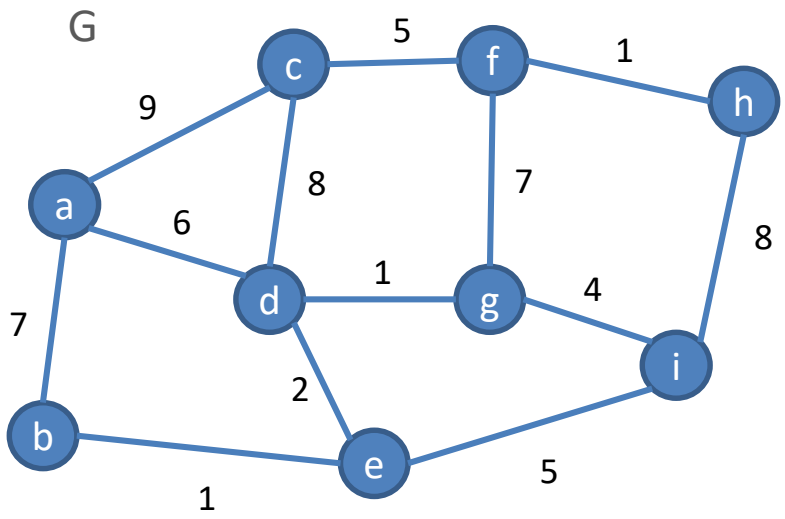
Dessa forma são necessárias pelo menos $n - 1$ iterações (em que n representa o número vértices do grafo). Contudo, pode ser necessário realizar mais iterações devido a necessidade de se ignorar arestas que formem ciclos.

O raciocínio está voltado para a formação da árvore a partir da inclusão de arestas, e não de vértices, como no algoritmo de Prim.

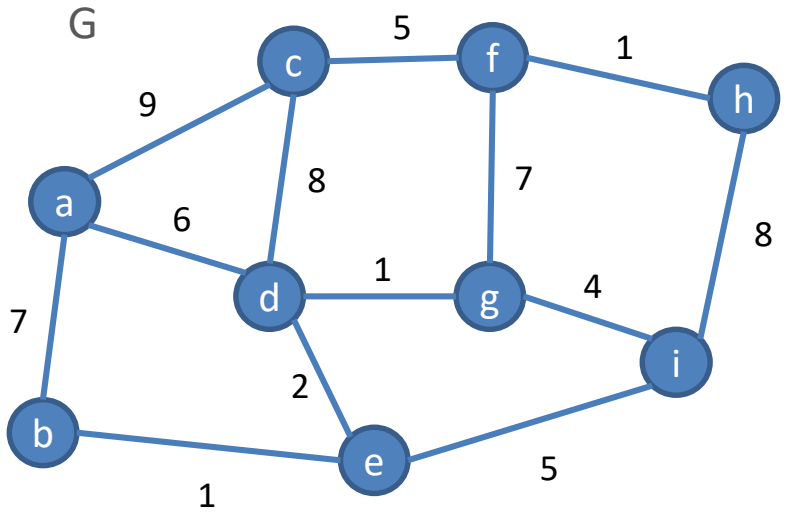
Método de Kruskal – Algoritmo

1. Ordenar arestas em ordem não decrescente $e_1, e_2, e_3, \ldots$
2. $V(T) \leftarrow V(G)$; // Adiciona todos vértices à AGM
3. $E(T) \leftarrow \{ e_1 \}$; // Inicializar arestas da AGM
4. $j \leftarrow 2$; // Indica aresta a ser analisada
5. <u>enquanto</u> $|E(T)| < |V(T)| - 1$ <u>efetuar</u> // Se ainda não for conexo?
   a. <u>se</u> aresta e_j não forma ciclo com as arestas em E(T) <u>então</u>
      i. Acrescentar $\{v, w\}$ a E(T) // Adicionar nova aresta à AGM
   b. $j \leftarrow j + 1$; // Indica próxima aresta

Método de Kruskal – Exemplo

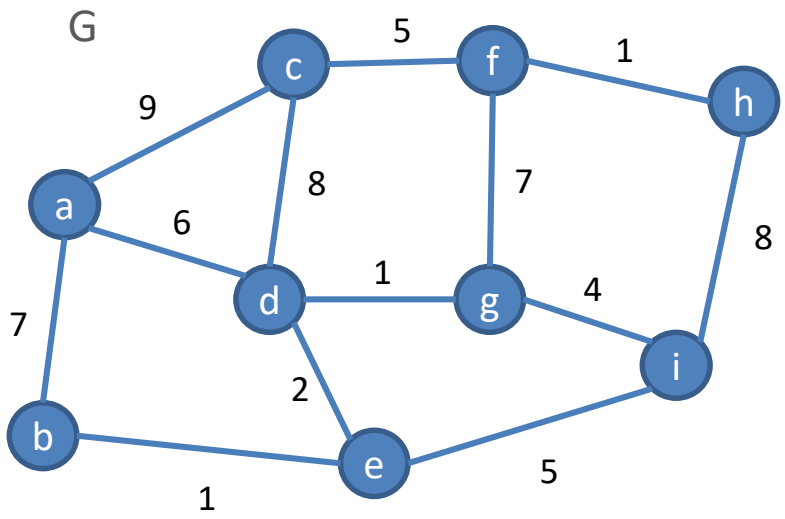


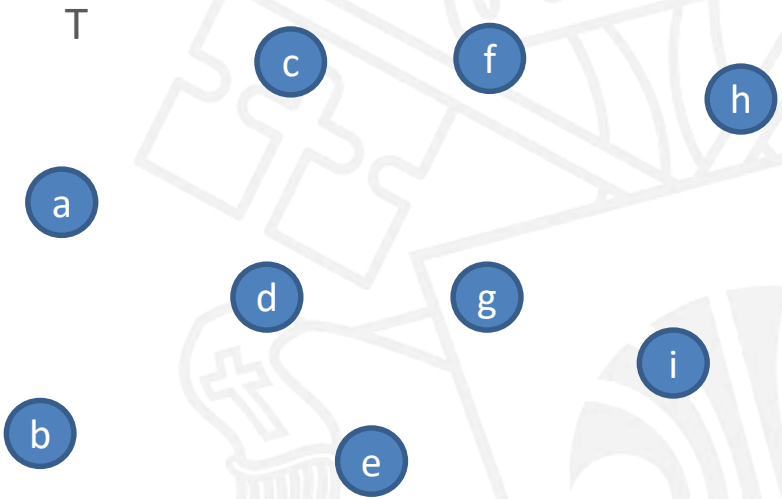
Método de Kruskal – Exemplo



Lista ordenada de arestas: {b,e}, {d,g}, {f,h}, {d,e}, {g,i}, {e,i},
{c,f}, {a,d}, {a,b}, {f,g}, {c,d}, {h,i}, {a,c}

Método de Kruskal – Exemplo





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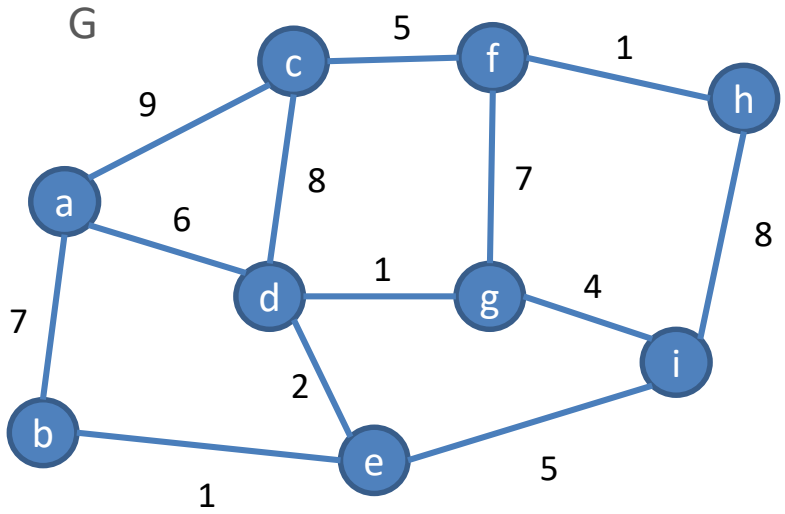
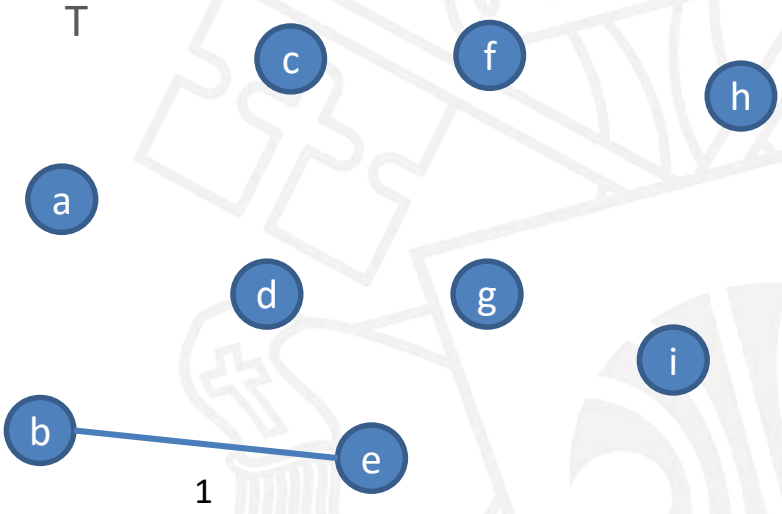
Método de Kruskal – Exemplo





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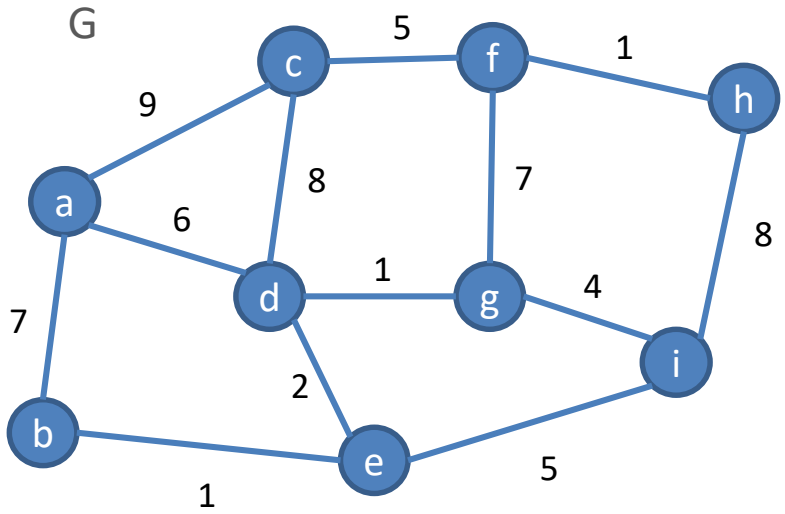
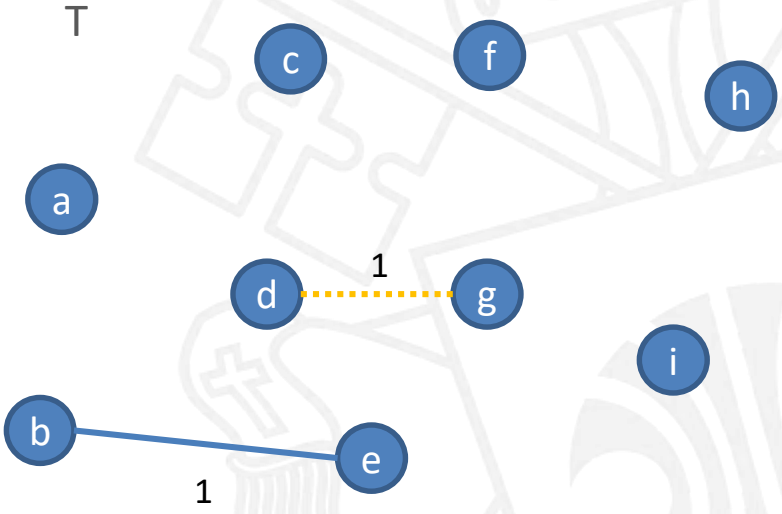
Método de Kruskal – Exemplo





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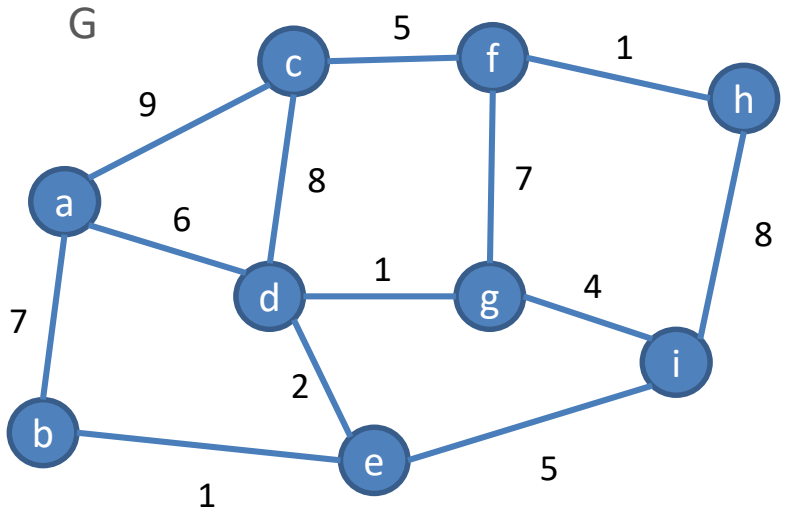
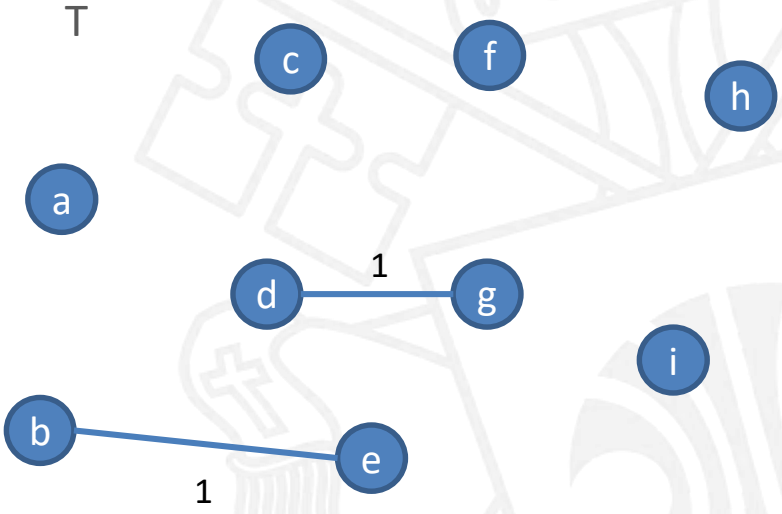
Método de Kruskal – Exemplo





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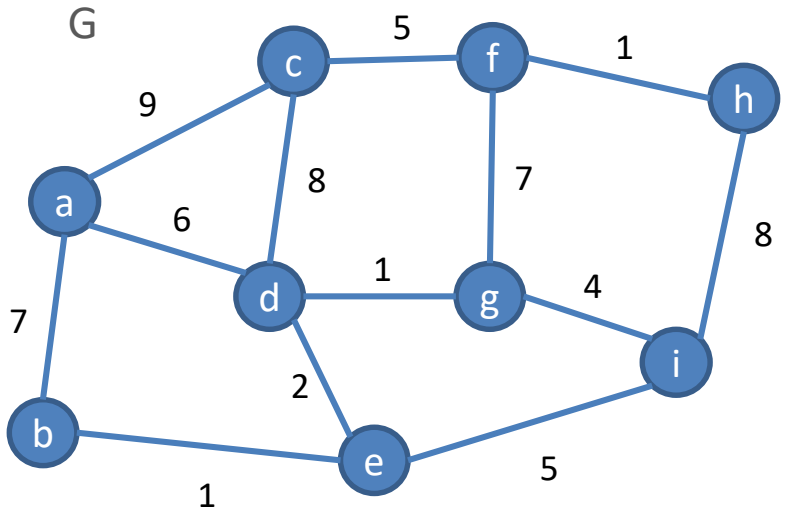
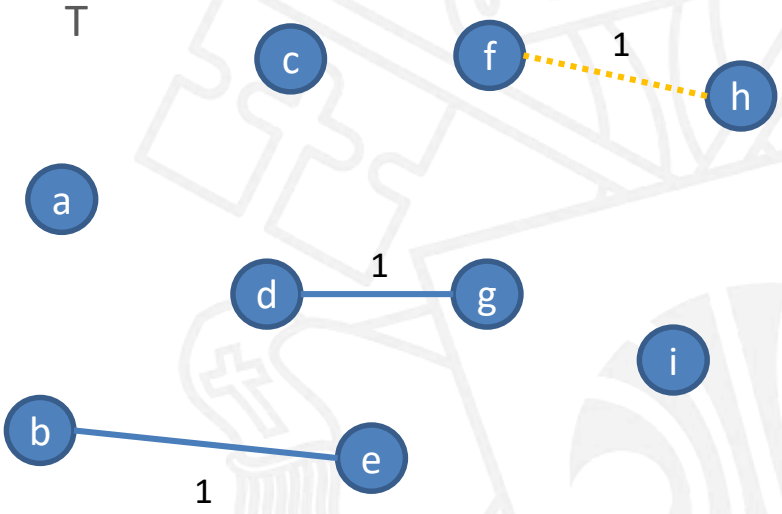
Método de Kruskal – Exemplo





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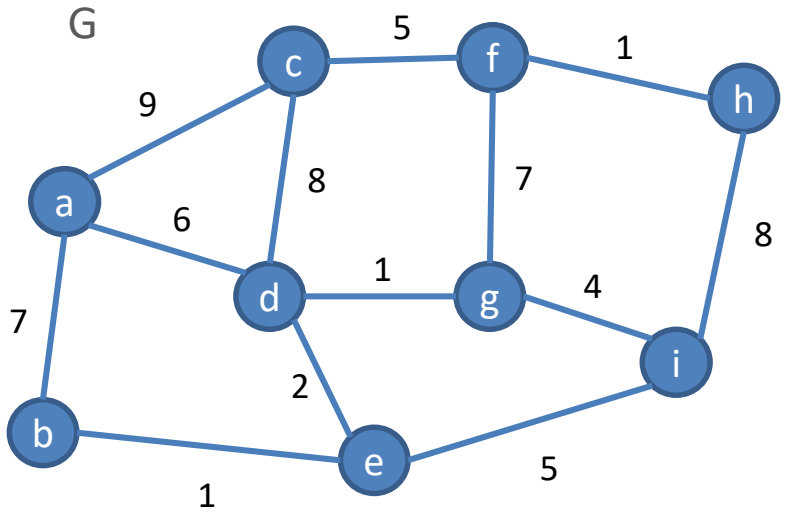
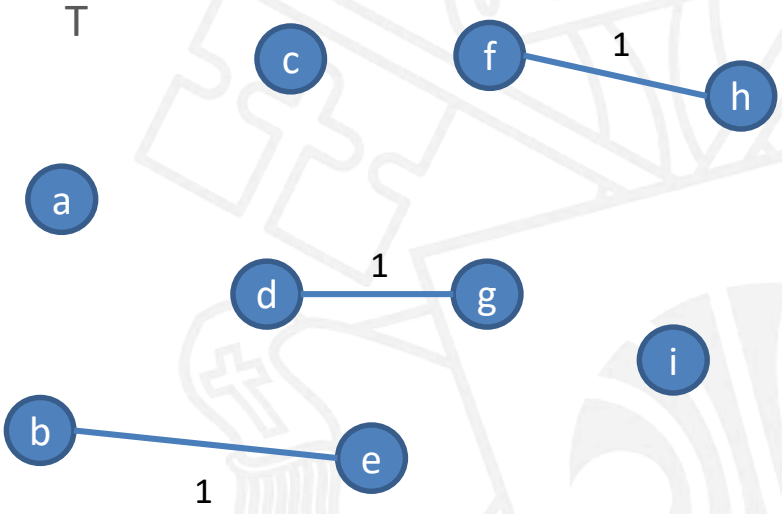
Método de Kruskal – Exemplo





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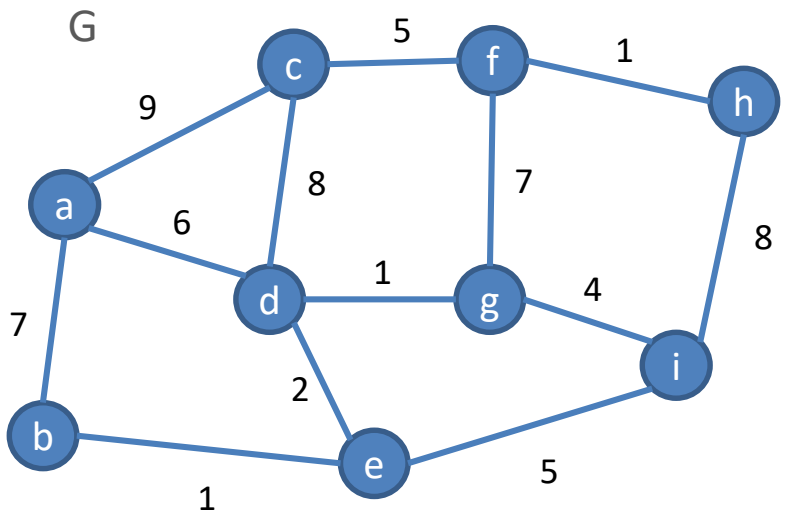
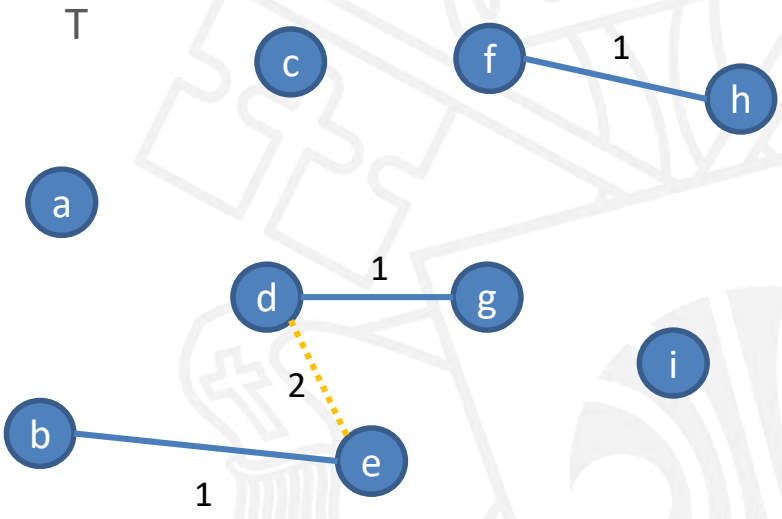
Método de Kruskal – Exemplo





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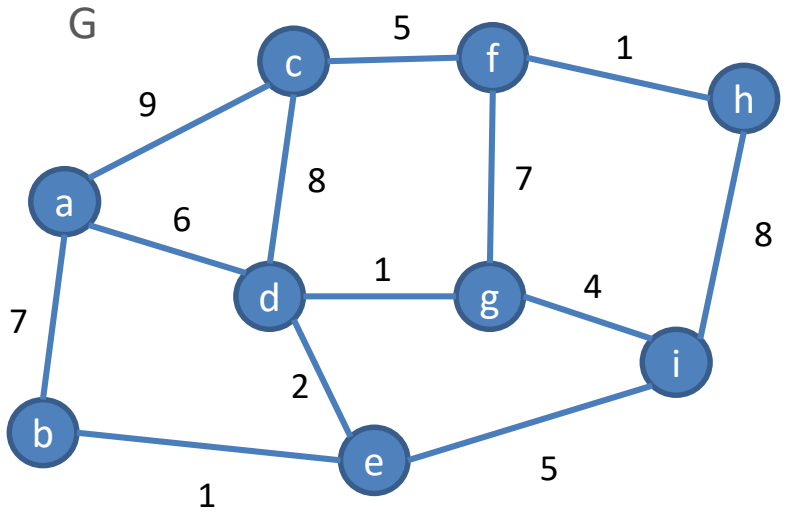
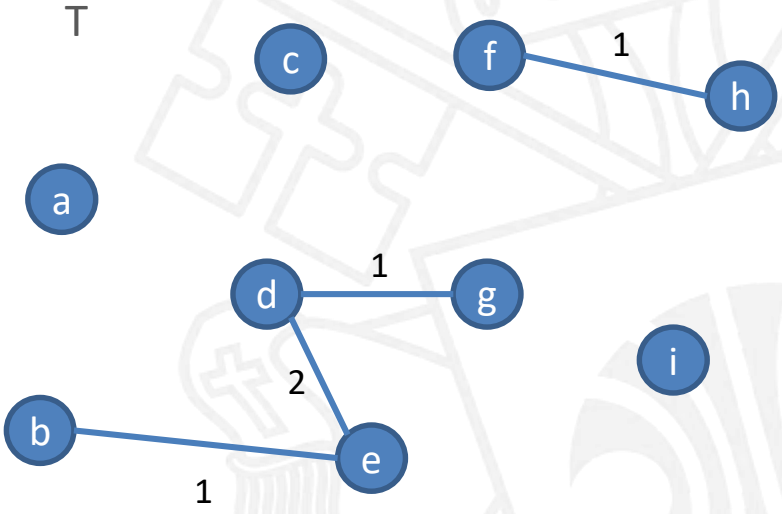
Método de Kruskal – Exemplo





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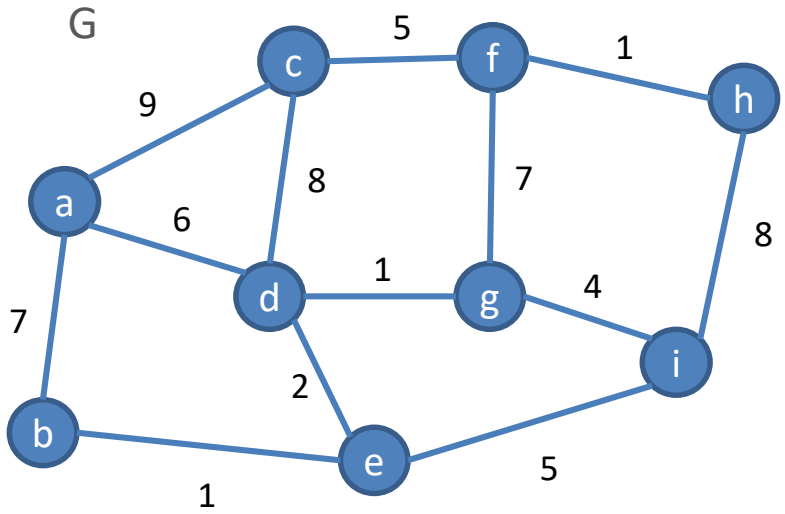

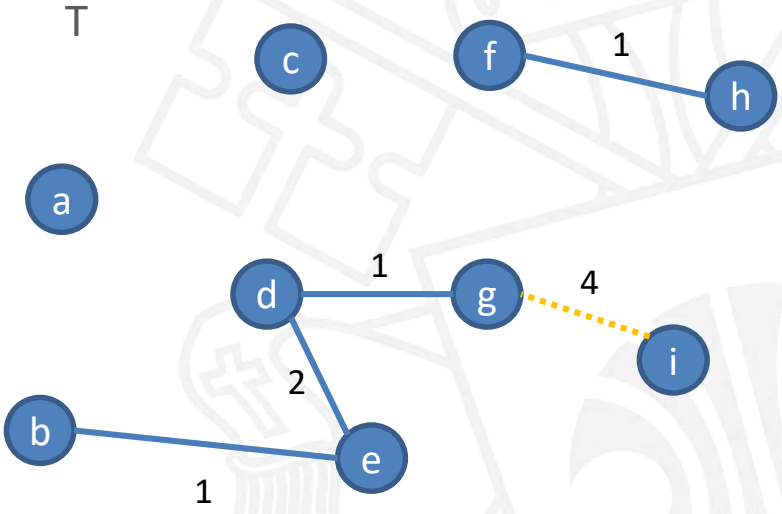

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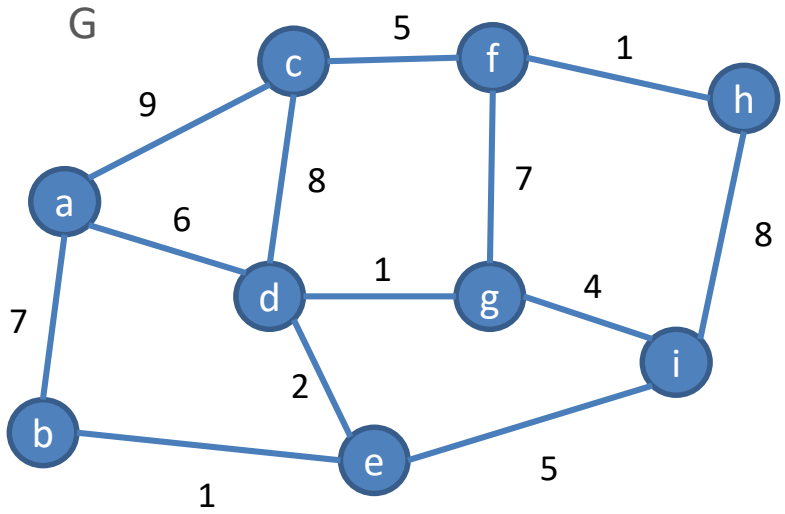
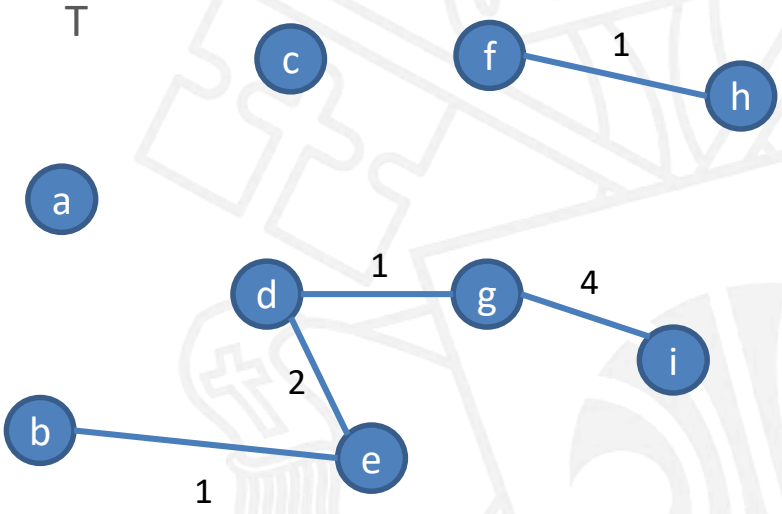
Método de Kruskal – Exemplo





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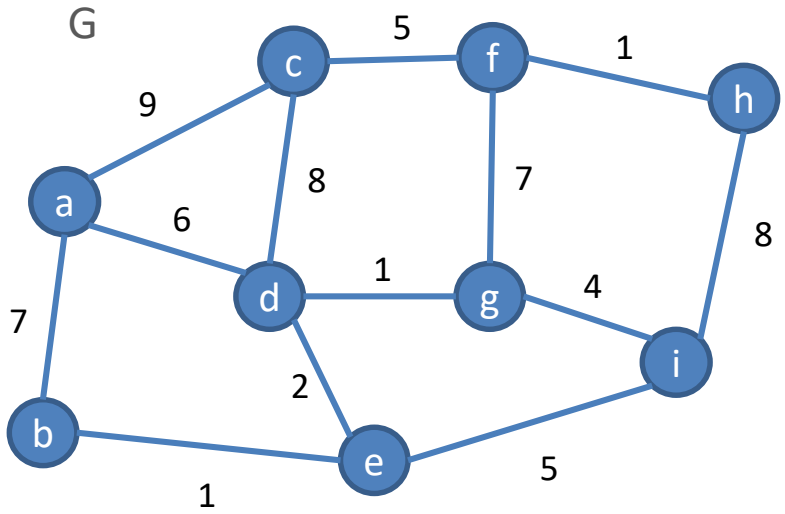
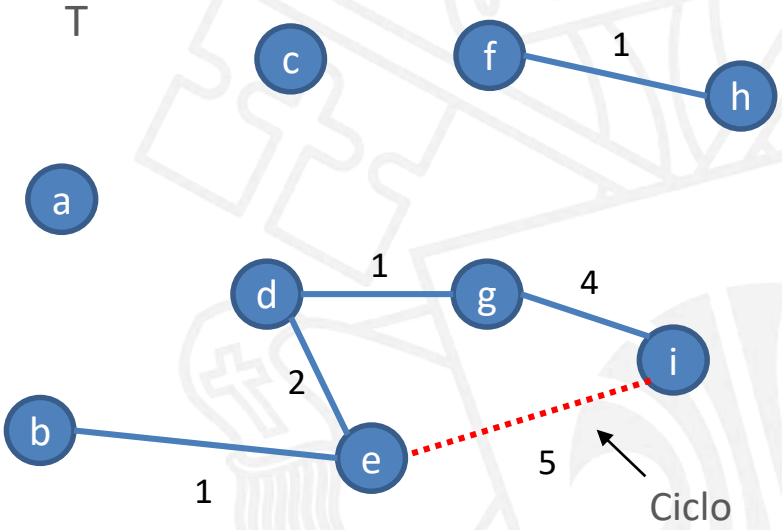
Método de Kruskal – Exemplo





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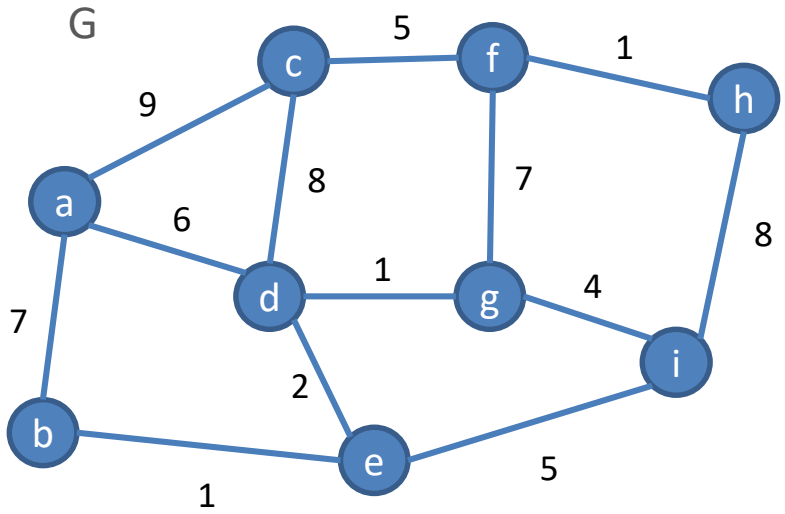
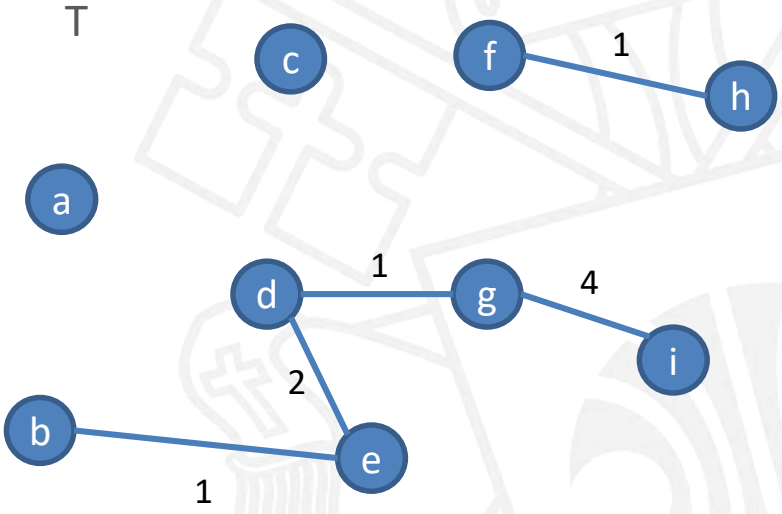
Método de Kruskal – Exemplo





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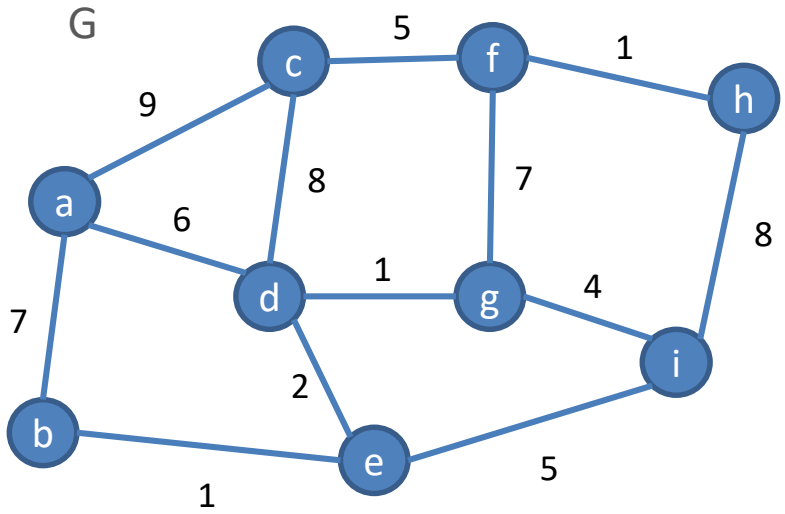
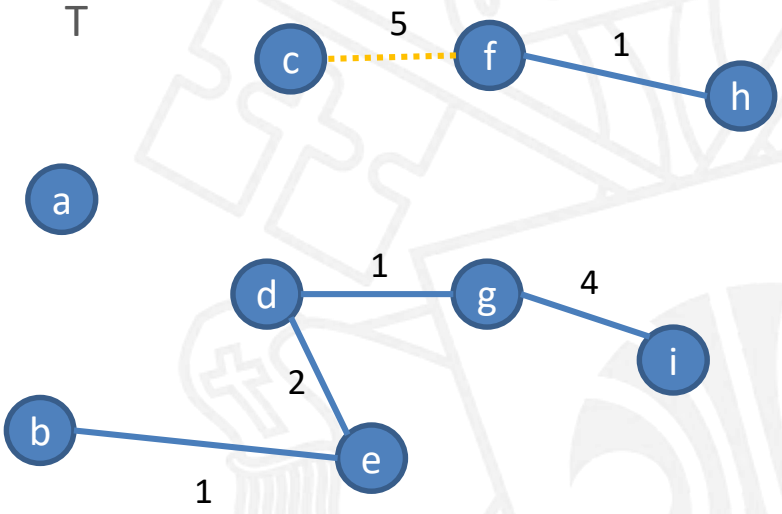
Método de Kruskal – Exemplo





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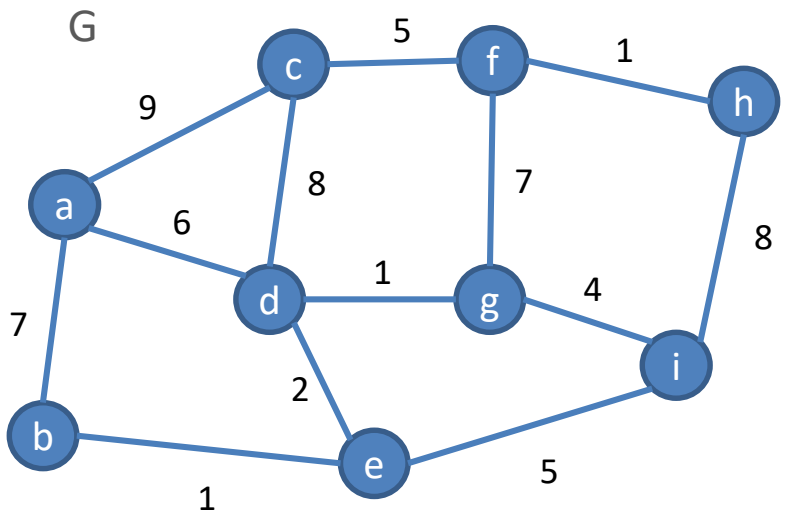
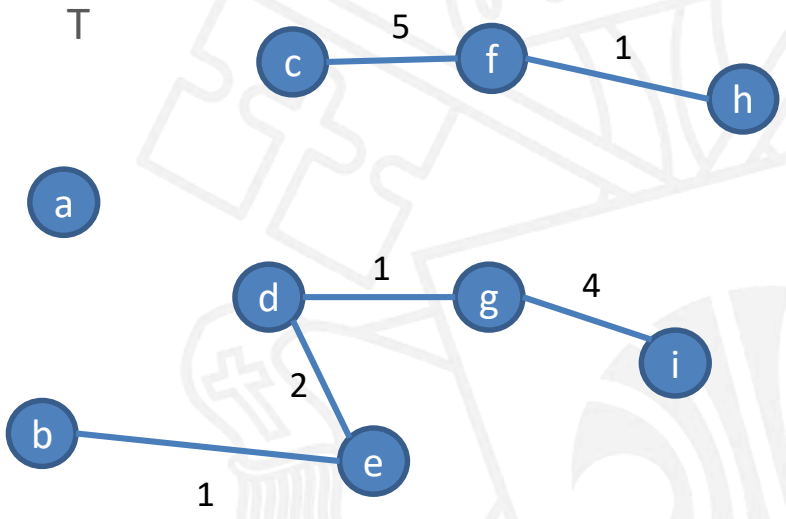
Método de Kruskal – Exemplo





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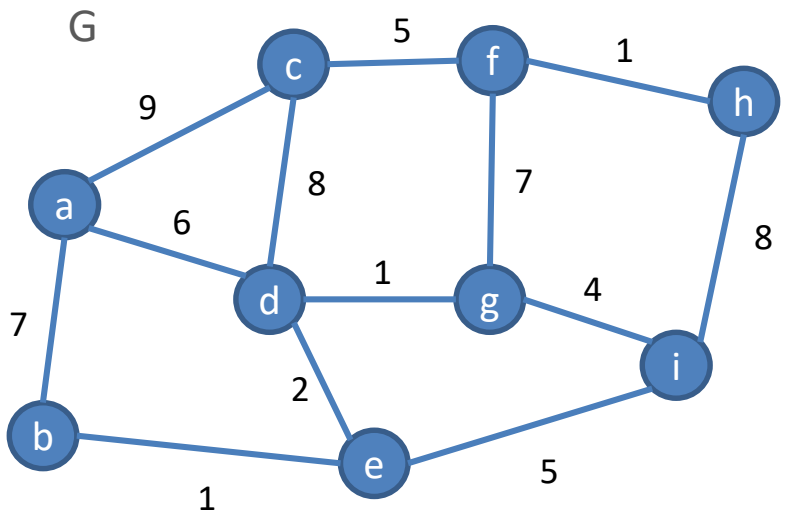
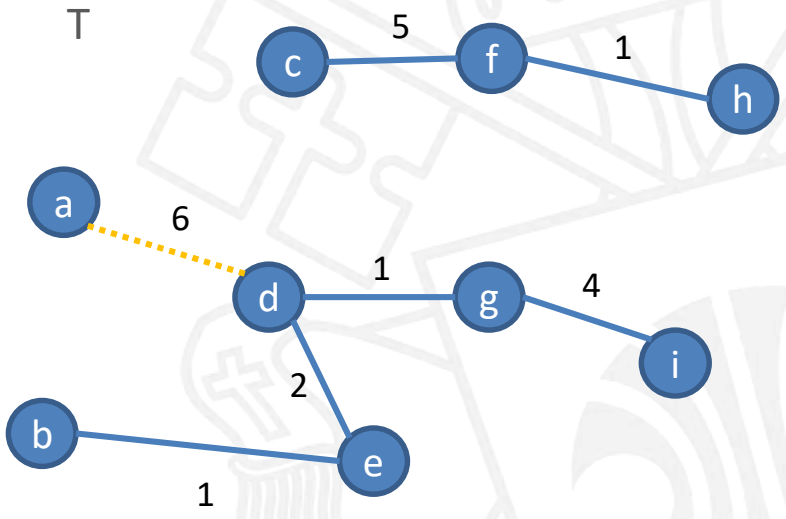
Método de Kruskal – Exemplo





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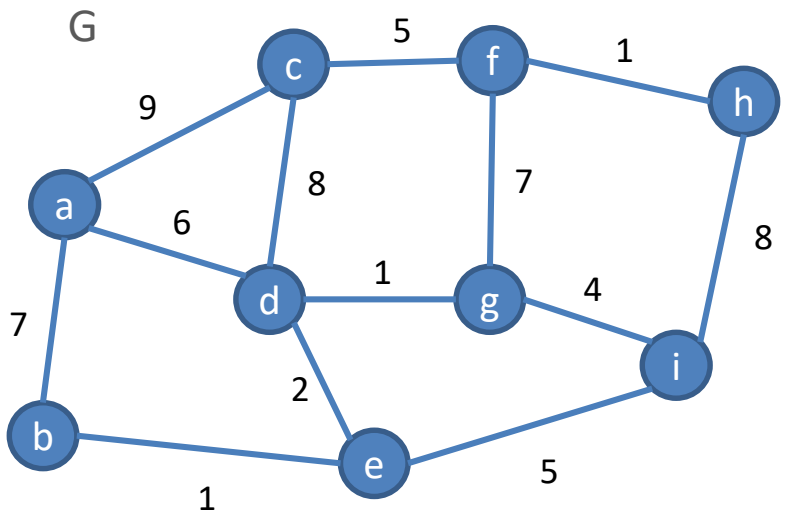
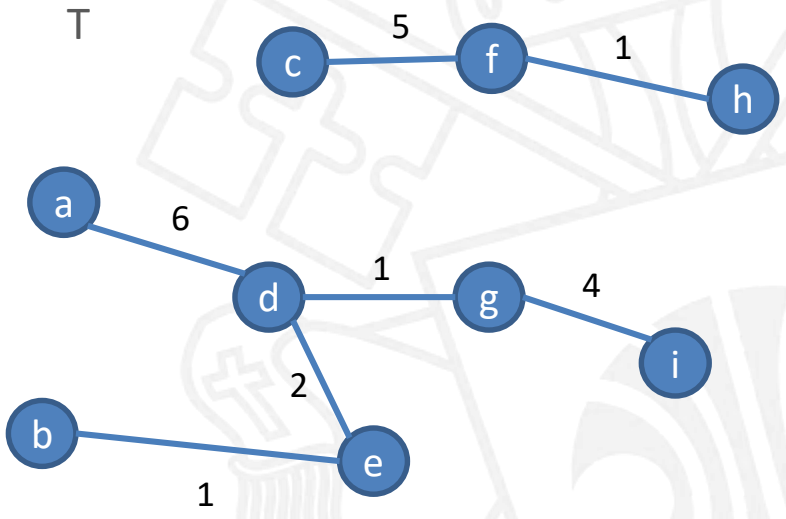
Método de Kruskal – Exemplo





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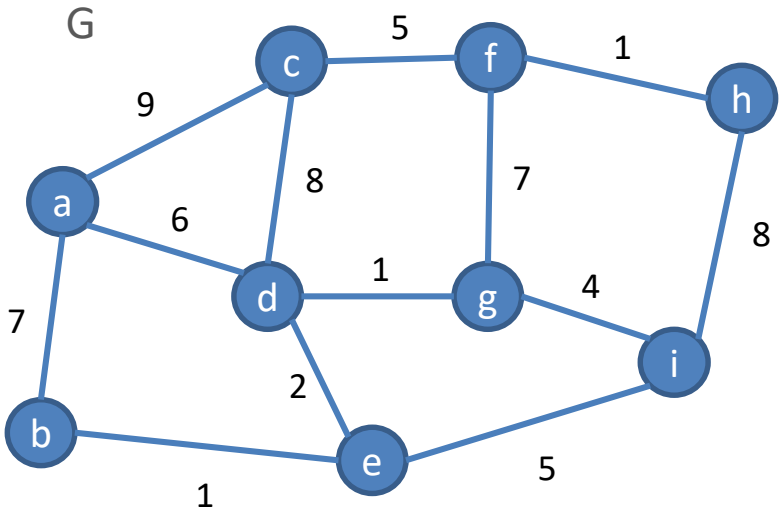

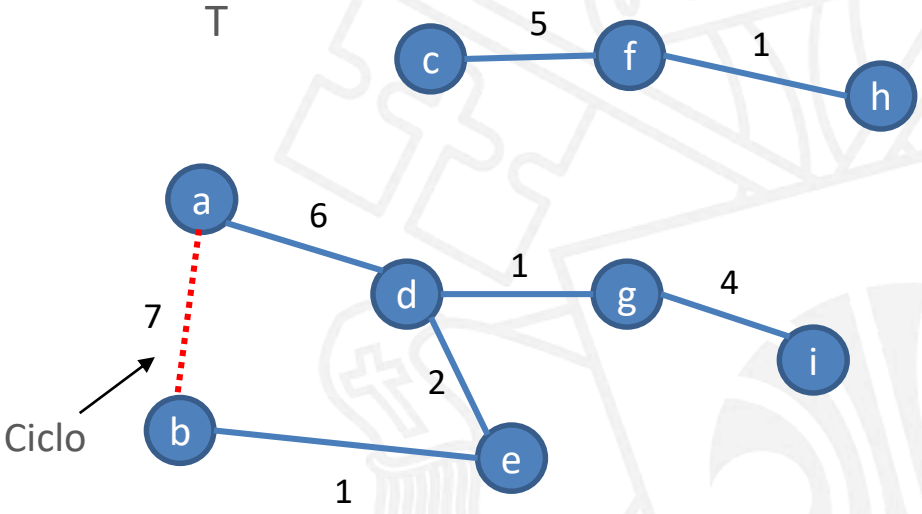

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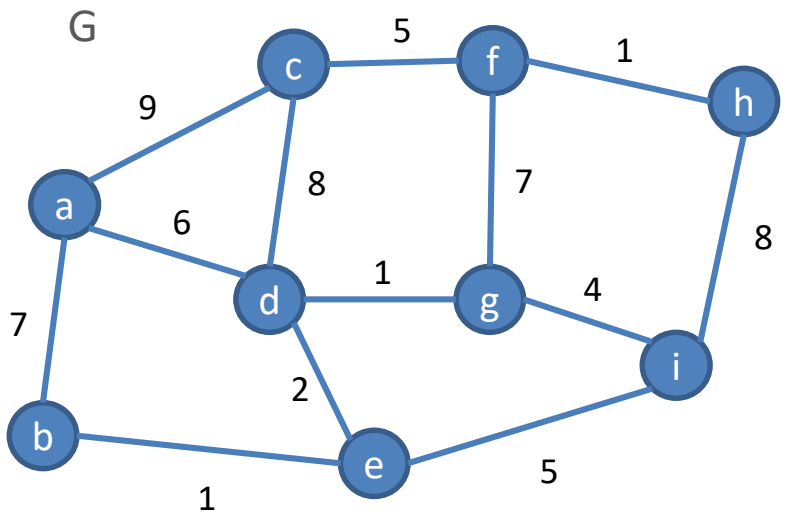
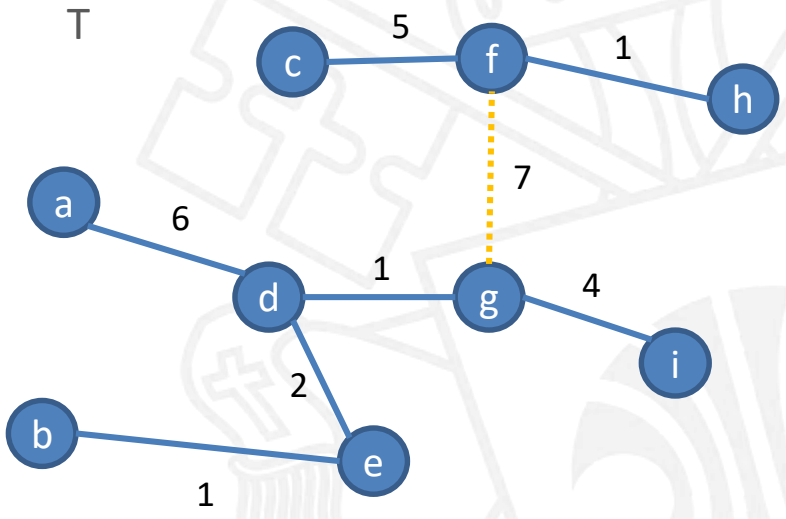
Método de Kruskal – Exemplo





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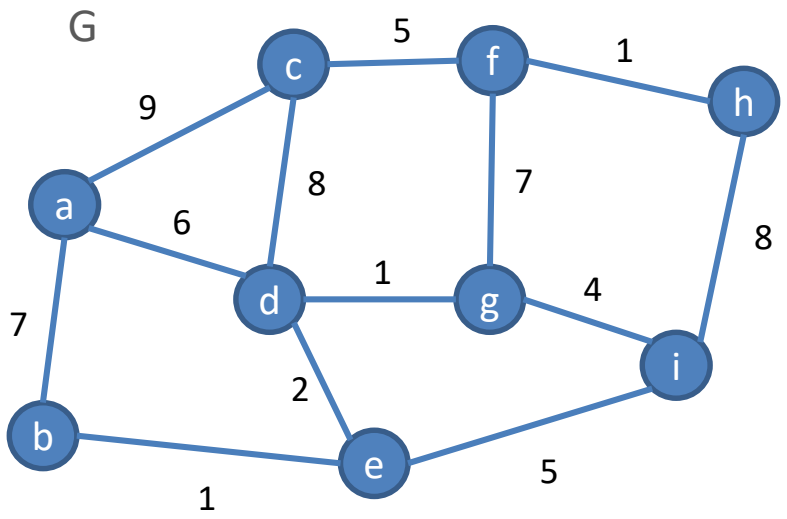
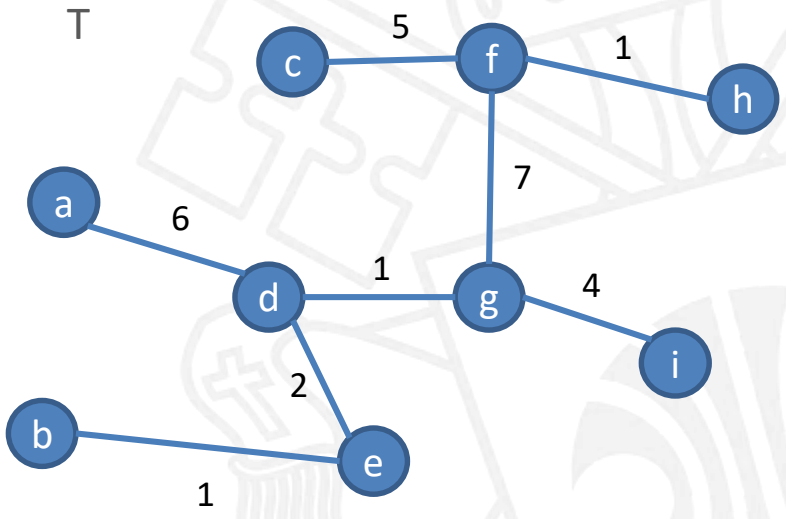
Método de Kruskal – Exemplo





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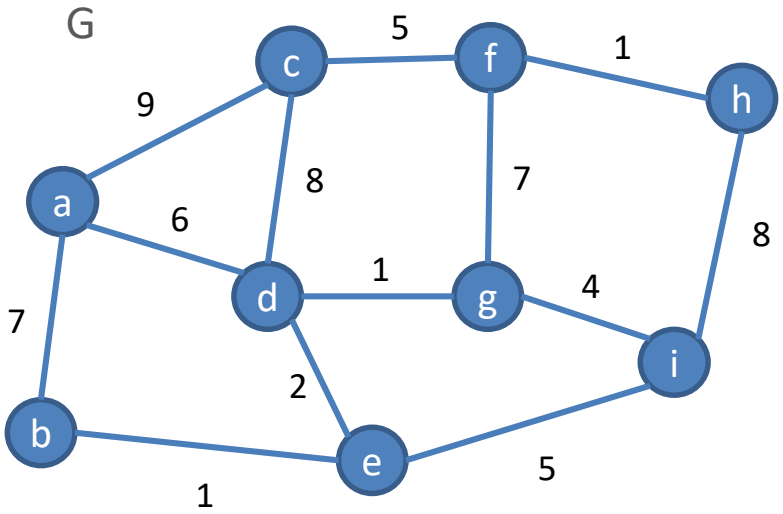
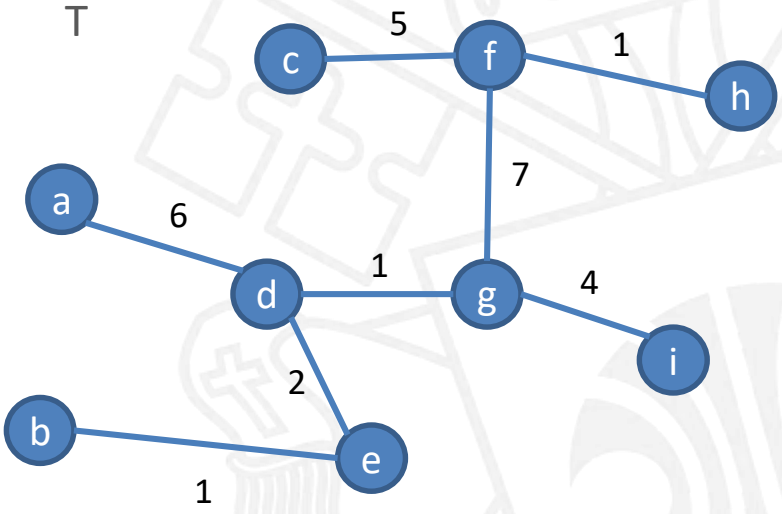
Método de Kruskal – Exemplo





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